

Tissue and cellular tropism, pathology and pathogenesis of Ebola and Marburg viruses

Roosecelis Brasil Martines,¹ Dianna L Ng,¹ Patricia W Greer,¹ Pierre E Rollin² and Sherif R Zaki^{1*}

¹ Infectious Diseases Pathology Branch, Division of High Consequence Pathogens and Pathology, Centers for Disease Control and Prevention, Atlanta, GA, USA

² Viral Special Pathogens Branch, Division of High Consequence Pathogens and Pathology, Centers for Disease Control and Prevention, Atlanta, GA, USA

*Correspondence to: Sherif R Zaki, Infectious Diseases Pathology Branch, Division of High Consequence Pathogens and Pathology, NCEZID, CDC, 1600 Clifton Road NE, MS G32, Atlanta, GA 30329-4018, USA. E-mail: szaki@cdc.gov

Abstract

Ebola viruses and Marburg viruses include some of the most virulent and fatal pathogens known to humans. These viruses cause severe haemorrhagic fevers, with case fatality rates in the range 25–90%. The diagnosis of filovirus using formalin-fixed tissues from fatal cases poses a significant challenge. The most characteristic histopathological findings are seen in the liver; however, the findings overlap with many other viral and non-viral haemorrhagic diseases. The need to distinguish filovirus infections from other haemorrhagic fevers, particularly in areas with multiple endemic viral haemorrhagic agents, is of paramount importance. In this review we discuss the current state of knowledge of filovirus infections and their pathogenesis, including histopathological findings, epidemiology, modes of transmission and filovirus entry and spread within host organisms. The pathogenesis of filovirus infections is complex and involves activation of the mononuclear phagocytic system, with release of pro-inflammatory cytokines, chemokines and growth factors, endothelial dysfunction, alterations of the innate and adaptive immune systems, direct organ and endothelial damage from unrestricted viral replication late in infection, and coagulopathy. Although our understanding of the pathogenesis of filovirus infections has rapidly increased in the past few years, many questions remain unanswered.

Copyright © 2014 Pathological Society of Great Britain and Ireland. Published by John Wiley & Sons, Ltd.

This article has been contributed to by US Government employees and their work is in the public domain in the USA

Keywords: filovirus; pathology; immunohistochemistry; Ebola virus; Marburg virus; pathogenesis; haemorrhagic fever; electron microscopy; autopsy

Received 22 September 2014; Revised 30 September 2014; Accepted 1 October 2014

Introduction

Viral haemorrhagic fevers (VHFs) is a term used for severe systemic infections characterized by fever and haemorrhage that are caused by four distinct families of small, lipid-enveloped RNA viruses, including Arenaviridae, Filoviridae, Bunyaviridae and Flaviviridae. All are negative-stranded viruses, with the exception of flaviviruses, which are positive-stranded. Each of these viruses is maintained in its own particular transmission cycle, which may involve non-human primates (NHPs), bats, rodents, domestic ruminants, humans, mosquitoes and ticks [1,2].

The family Filoviridae consists of Marburg viruses and Ebola viruses [3]. Since the discovery of *Marburgvirus* (MARV) in 1967 and *Ebolavirus* (EBOV) in 1976, these viruses have traditionally been associated with sporadic outbreaks, predominantly in geographically remote areas of Central Africa [4]. The largest recorded Marburg outbreak to date occurred in Angola in 2004–2005, with 227 reported deaths [5,6]. At the

time of writing this article, the largest reported outbreak of EBOV is ongoing and is the first outbreak ever to occur in West Africa, affecting Guinea, Liberia, Sierra Leone and Nigeria. As of 31 August 2014, there have been 3707 suspected and confirmed cases, with 1848 deaths [7,8]. Without additional interventions, the projected estimate of total cases in Liberia and Sierra Leone by 20 January 2015 is approximately 550 000 cases, or 1.4 million when corrected for under-reporting [9].

Pathology has traditionally played a critical role in the discovery and advancement of our knowledge of emerging infectious diseases [10–12]. Autopsies can determine the pathological features of the disease, help identify the causative pathogen and provide new insights into the pathogenesis of infections. Filoviruses are highly virulent and are classified as Biosafety Level (BSL)-4 pathogens, that is, they are potentially transmissible via aerosols and cause life-threatening disease for which there is no vaccine or therapy [13]. Biosafety is critical for autopsy personnel, particularly when dealing with infectious agents. Standard precautions for infection

control, as outlined by the Centers for Disease Control and Prevention (CDC), integrates hand hygiene, use of personal protective equipment – such as gloves, gowns, masks, respirators, goggles and face shields – safe injection practices, safe handling of potentially contaminated equipment or surfaces, and respiratory hygiene [14]. The World Health Organization recommends that post-mortem examinations of VHF patients should be performed by trained personnel and limited to essential evaluations only [15]. The 2014 CDC recommendation for Ebola patients in US hospitals and mortuaries is that autopsies should be avoided and that, if an autopsy is necessary, the state health department and CDC should be consulted regarding additional precautions [16].

Numerous reports using animal models and *in vitro* studies to explore the pathogenesis of filoviruses have been published. These have made significant strides in our understanding of filoviruses and will be referenced throughout the article. On the other hand, human studies have been relatively few because of: (a) biosafety concerns; (b) occurrence of the disease in remote areas with little modern infrastructure; (c) sporadic and unpredictable timing of outbreaks; and (d) poor filovirus surveillance. Furthermore, efforts to explain the pathogenesis of Ebola and Marburg haemorrhagic fevers must consider a multitude of factors, including clinicopathological and epidemiological data. We review how experimental animal and human pathology studies have contributed to a better understanding of filovirus pathogenesis and to our ability to diagnose filovirus infections. This review will also attempt to identify the gaps in our knowledge that represent critical challenges for the future.

The agents

Filoviruses are filamentous, enveloped, single-stranded, negative-sense RNA viruses. The family *Filoviridae* includes two distinct genera, *Marburgvirus* and *Ebolavirus*, which differ in their glycoproteins at both the nucleotide and amino acid levels [3,17]. Their genomes have a length of approximately 19 kb, which encodes seven structural proteins (3'-UTR-NP-VP35-VP40-GP-VP30-VP24-L-5'-UTR) and one non-structural protein. These pleomorphic filamentous viruses enter the host cell through membrane-embedded glycoproteins and then replicate, using seven genes encoded in their negative-sense RNA genome. The genus *Ebolavirus* includes five species, three of which are highly pathogenic in humans: *Zaire Ebolavirus* has case fatality rates (CFRs) in the range of 70–90%; *Sudan Ebolavirus*, CFR of about 50%; and *Bundibugyo Ebolavirus*, CFR of about 25% [2,3,18]. Only one case of *Tai Forest Ebolavirus* infection has been reported [19], from which the patient recovered. Although *Reston Ebolavirus* causes disease in non-human primates and pigs, it appears to be non-pathogenic in humans [20]. The genus *Marburgvirus* contains only one species of

virus and has a variable CFR, eg a CFR of 25% was noted in 1967 in Germany, while CFRs of 90% were seen in more recent outbreaks in Africa [5,21,22].

Transmission, ecology and epidemiology

Filoviruses are zoonotic diseases transmitted through direct contact with infected live or dead animals. The main means of transmission in humans is contact with the blood, secretions or body fluids of an infected human, non-human primate or reservoir host animal (Figure 1A, B) [7,20,23–25]. Humans can transmit the virus as soon as they are febrile and continue to be infectious during the later stages of the disease as well as after death, eg during ritual body washing [6,7,22,26,27]. Rare human infections have been documented associated with accidental laboratory exposure or during conduction of a necropsy (Figure 2) [8,19,28,29].

Several studies have attempted to identify the natural reservoir of filoviruses. Epidemiological data from numerous Ebola outbreaks, in combination with molecular and virological analyses, suggest that fruit bats, including *Hypsignathus monstrosus*, *Epomops franqueti*, *Rousettus aegyptiacus* and *Myonycteris torquata*, are likely to be reservoirs of these viruses [25,30–33] (Figure 1A). On the other hand, *Rousettus aegyptiacus*, fruit bats of the family Pteropodidae family, have been definitively identified as the natural reservoir of Marburg virus (Figure 1B) [34,35]. In several cases, epidemiological evidence suggests a link between bats and Marburg virus infection, and includes a French engineer who became infected by Marburg virus after visiting caves with large bat populations in Kenya in 1980 [24], and four miners in Uganda in 2007 who likely acquired Marburg virus through exposure to bats or bat secretions in a mine [36]. However, many questions regarding the mechanism by which bats shed and transmit the viruses remain unanswered.

Since the discovery of Marburg virus in 1967, there have been 11 documented outbreaks of Marburg virus; since the discovery of Ebola virus in 1976, there have been 24 Ebola outbreaks, confirmed by virus isolation or by molecular and biological methods [7,20,29,37,38]. A summary of all known filoviruses outbreaks and CFRs can be found in Table 1. Reston virus was introduced into quarantine facilities in the USA and Italy in 1989, 1990, 1992 and 1996 from non-human primates imported from primate facilities in The Philippines. These occurrences were linked to Reston virus isolated from primate export facilities in The Philippines. In 2008, pigs were identified as a host for Reston virus in The Philippines [39]. There have been 13 asymptomatic and serologically confirmed human cases associated with these different events since 1989. Geographical distribution of Marburg virus and Ebola virus outbreaks since 1967–2014 are illustrated in Figure 2.

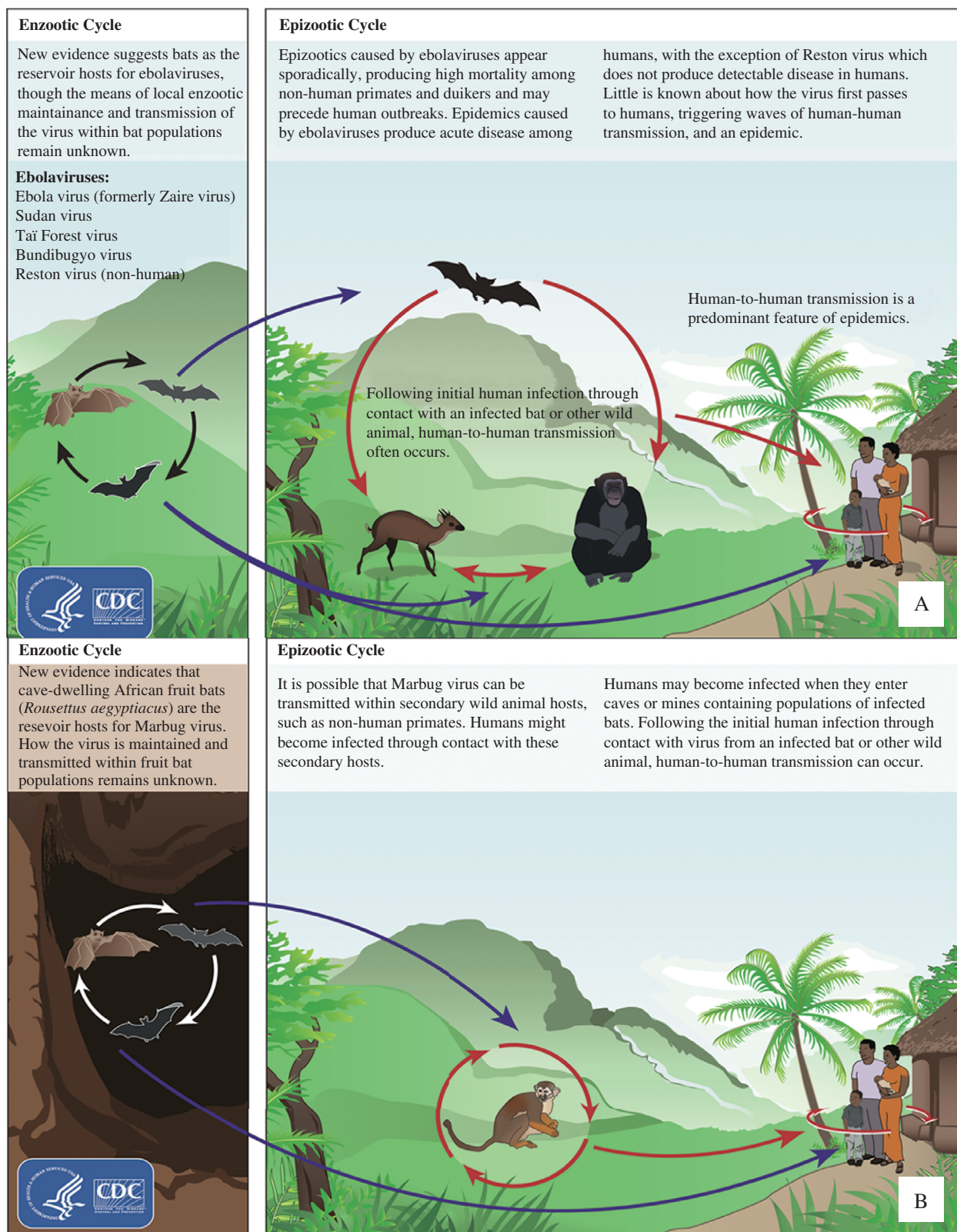


Figure 1. Filovirus ecology. Illustrations detailing both enzootic and epizootic cycles of filoviruses and transmission to humans, courtesy of Craig Manning, Viral Special Pathogens Branch, and Alissa Eckert, Medical Illustrator, CDC

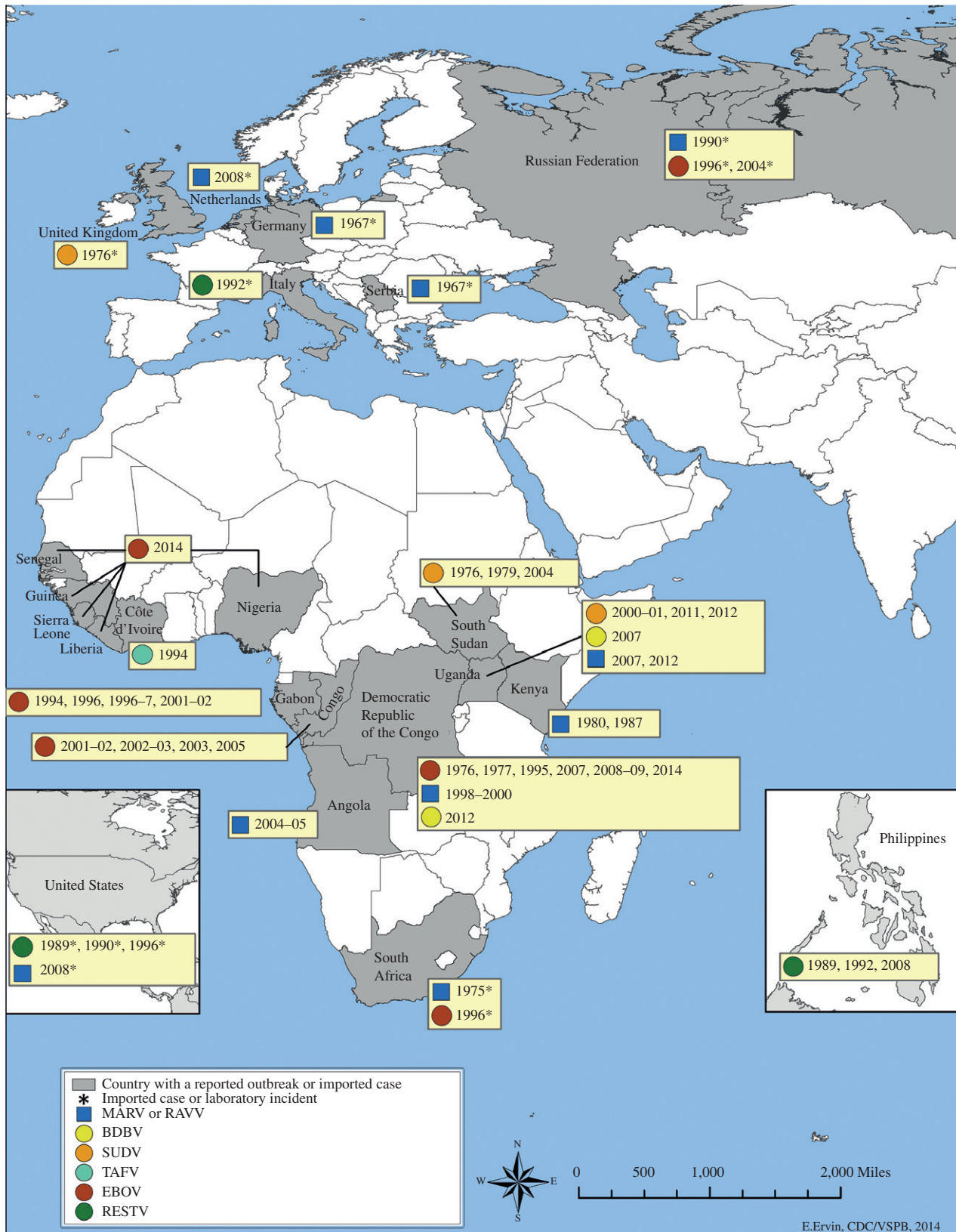


Figure 2. Worldwide geographic distribution of filovirus haemorrhagic fever cases, 1967–2014. Illustration courtesy of Elizabeth Ervin, Viral Special Pathogens Branch, CDC

Table 1. Chronology and location of human filovirus outbreaks (1967–2014), including all laboratory-confirmed Ebola and Marburg haemorrhagic fever cases

Year(s)	Country	Number of human cases	Fatal cases and CFR	Filovirus
1967	Germany and Yugoslavia	31	7 (23%)	MARV
1975	Johannesburg, South Africa	3	1 (33%)	MARV
1976	Zaire (Democratic Republic of the Congo - DRC)	318	280 (88%)	EBOV
1976	Sudan (South Sudan)	284	151 (53%)	SUDV
1976	England	1	0 (0)	SUDV
1977	Zaire	1	1 (100%)	EBOV
1979	Sudan (South Sudan)	34	22 (65%)	SUDV
1980	Kenya	2	1 (50%)	MARV
1987	Kenya	1	1 (100%)	MARV
1990	Russia	1	1 (100%)	MARV
1994	Gabon	52	31 (60%)	TAFV
1994	Ivory Coast	1	0 (0)	EBOV
1995	DRC (formerly Zaire)	315	250 (81%)	EBOV
1996	Gabon	37	21 (57%)	EBOV
1996–1997	Gabon	60	45 (74%)	EBOV
1996	South Africa	2	1 (50%)	EBOV
1996	Russia	1	1 (100%)	EBOV
1998–2000	DRC	154	128 (83%)	MARV
2000–2001	Uganda	425	224 (53%)	SUDV
October 2001–March 2002	Gabon	65	53 (82%)	EBOV
October 2001–March 2002	Republic of Congo	57	43 (75%)	EBOV
December 2002–April 2003	Republic of Congo	143	128 (89%)	EBOV
November–December 2003	Republic of Congo	35	29 (83%)	EBOV
2004	Sudan (South Sudan)	17	7 (41%)	SUDV
2004–2005	Angola	252	227 (90%)	MARV
2004	Russia	1	1 (100%)	EBOV
2007	Uganda	4	1 (25%)	MARV
2007	DRC	264	187 (71%)	EBOV
December 2007–January 2008	Uganda	149	37 (25%)	BDBV
2008	USA ex Uganda	1	0 (0)	MARV
2008	Netherlands ex Uganda	1	1 (100%)	MARV
December 2008–February 2009	DRC	32	15 (47%)	EBOV
May 2011	Uganda	1	1 (100%)	SUDV
2012	Uganda	15	4 (27%)	MARV
June–October 2012	Uganda	11*	4* (36.4%)	SUDV
June–November 2012	DRC	36*	13* (36.1%)	BDBV
November 2012–January 2013	Uganda	6*	3* (50%)	SUDV
March 2014–present	Guinea, Liberia, Sierra Leone, Nigeria	3017*	1513 (50%)*	EBOV

Modified from [5,7], courtesy of the Centers for Disease Control and Prevention.

DRC, Democratic Republic of the Congo; MARV, Marburg virus; EBOV, Ebola virus; SUDV, Sudan virus; TAFV, Tai forest virus; BDBV, Bundibugyo virus.

*Numbers reflect laboratory-confirmed cases only.

Clinical presentation and laboratory diagnosis

Early on in the clinical course, the presentation of filovirus-infected patients is non-specific and difficult to distinguish from other endemic diseases, such as Lassa fever, malaria, cholera or typhoid fever. Patients can also develop simultaneous infections with other pathogens, thereby further complicating the diagnosis [38,40,41]. The asymptomatic incubation period of filoviruses is 2–21 days. The disease manifests abruptly with non-specific flu-like symptoms, including chills, fever, myalgia and general malaise. This is followed by lethargy, nausea, vomiting, abdominal pain, anorexia, diarrhoea, coughing, headache and hypotension. Haemorrhagic manifestations vary in severity and are seen in 30–50% of patients. These include a maculopapular rash and mucosal bleeding, especially in the gastrointestinal and genitourinary tracts [6,7,29,31].

In severe and fatal forms, death occurs 6–16 days after the onset of symptoms as a result of hypotensive shock and multi-organ failure, including hepatic damage and renal failure [29,40]. Features of illness seen in fatal cases include a high viral load in blood, a marked rise in blood neutrophil count, a fall in lymphocytes and platelets, and coagulopathy [42,43].

Several diagnostic methods to detect filoviruses are available, including reverse-transcription polymerase chain reaction (RT–PCR), serology, virus isolation, electron microscopy (EM), histopathology and immunohistochemistry (IHC) (Table 2, Figure 3A). Because of the biosafety hazards associated with filoviruses, these assays can be performed in only a few specialized laboratories worldwide. Such testing requires that potentially dangerous biological specimens be transported to these laboratories from remote sites, necessitating the preservation of a cold chain and rigorous international packaging and shipping procedures [5,44].

Table 2. Laboratory diagnostic arrays available at Centers for Disease Control and Prevention

Timeline of infection	Diagnostic tests available
Within a few days after symptoms begin	Antigen-capture enzyme-linked immunosorbent assay (ELISA) testing IgM ELISA Polymerase chain reaction (PCR) Virus isolation
Later in disease course or after recovery	IgM and IgG antibodies
Retrospectively in deceased patients	Histopathology Immunohistochemistry testing PCR Virus isolation Electron microscopy

Modified from [7], courtesy of the Centers for Disease Control and Prevention.

Ebola virus and Marburg virus are detectable in blood only after the onset of symptoms, most notably fever. Viral antigen and nucleic acid can be detected in blood from day 3 up to 7–16 days after onset of symptoms. IgM antibodies can appear as early as 2 days after the onset of symptoms and disappear 30–168 days after infection. Specific IgG antibodies can develop 6–18 days after onset and persist for up to 3–5 years [45]. Detection of filoviruses by PCR is available and is able to distinguish between the different viral variants in a range of tissues and fluid specimens [46]. Virus is generally detectable by real-time RT–PCR from 3–10 days after the onset of symptoms, and has even been detected several months after infection in certain secretions (Figure 3A) [7,26,46]. More recently, in outbreak settings, mobile laboratories commonly use PCR and/or ELISA analysis for rapid screening.

Ebola virus and Marburg virus antigens can be detected by IHC on formalin-fixed tissues by using specific polyclonal and monoclonal antibodies. These methods have a unique role in cases in which formalin-fixed tissues are the only specimens available for diagnostic testing. Formalin-fixed biopsy specimens are not infectious and may be sent without special precautions or refrigeration, and may be obtained in the most basic field conditions [44,47].

In situ hybridization (ISH) has been used in reports characterizing Ebola virus infections in animal models [48–50]. In cultures of alveolar macrophages, a notable increase in the amounts of EBOV RNA and protein in the cytoplasm of infected cells has been observed [51]. Experimental studies with guinea-pigs have detected macrophages containing EBOV RNA in lymph nodes, spleen, hepatocytes, Kupffer cells, circulating mononuclear cells, gastrointestinal tract, adrenal gland, kidney and urinary tract, lung, reproductive organs and heart [48].

Electron microscopy (EM) has long been used in the discovery and description of infectious diseases, particularly in cases where agents are unknown or unsuspected [52]. EM was contributory in elucidating the presence of a novel virus in the first outbreak of Marburg in Germany in 1967, in the first outbreak of Ebola virus in Zaire in 1976 [53–55] and in identifying the Reston virus infection of a monkey colony in Reston, VA, USA, in 1989 [56]. Several studies have demonstrated ultrastructural differences between MARV and EBOV, and have established EM to be a critical and definitive tool in the diagnosis of filoviruses (Figure 3B) [47,57,58].

Microarray analysis is a relatively newer research technique that was critical in the identification of pigs as a host for Reston virus in 2008. These pigs were

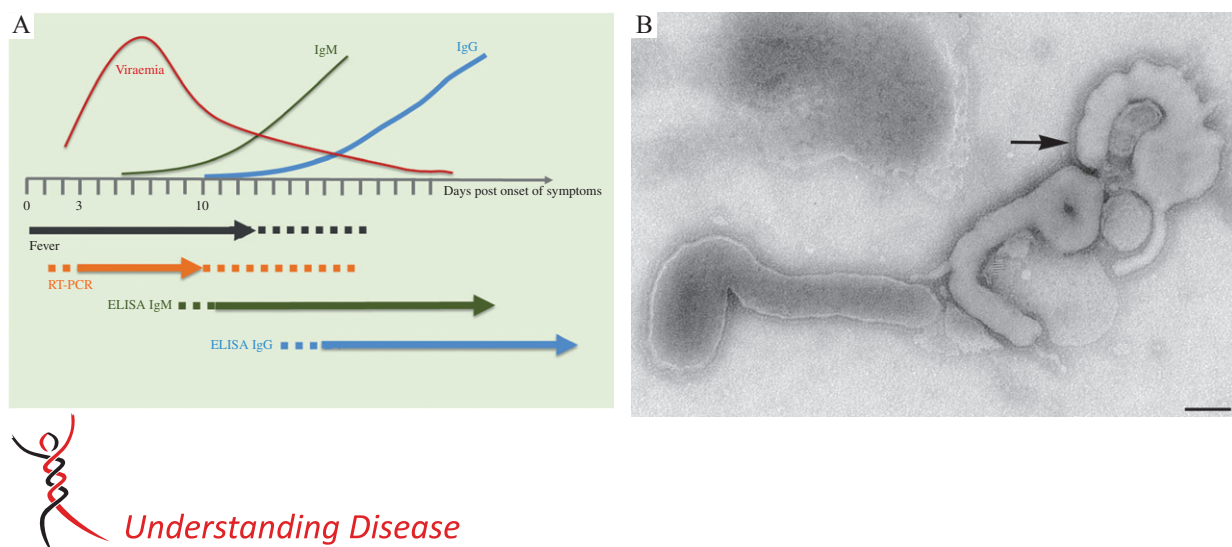


Figure 3. Laboratory diagnosis of filovirus haemorrhagic fever. (A) IgM antibodies can appear as early as 2 days after onset of symptoms and disappear after up to 3–6 months. Specific IgG antibodies can develop 6–18 days after the onset of symptoms and persist for several years. RT–PCR detects the virus generally 3–10 days after the onset of symptoms. (B) EM photomicrograph showing three Ebola virions, measuring between 500 nm and 1 µm in length; surface spikes composed of glycoprotein (arrow), 7–10 nm in length, can be seen extending from the virus surface; scale bar=0.1 µm. (A) Courtesy of Viral Special Pathogens Branch, CDC; (B) negative-stained EM courtesy of Maureen G Metcalfe, Infectious Disease Pathology Branch, CDC

co-infected with Porcine Reproductive and Respiratory Syndrome virus [39].

Pathology

Although more than 3000 fatal cases with documented filovirus infection have been recognized in the past 40 years, autopsies and post-mortem biopsies have been performed on a very small subset of these patients. This relatively small number can be partially attributed to filoviruses being category A pathogens, which have the potential to be transmitted to humans performing autopsies and necropsies [44,59,60]. Transmission may occur by percutaneous inoculation (ie injury) and by splashes to unprotected mucosa [44,61]. In contrast, aerosol transmission of filoviruses is highly controversial; a recent study in a BSL-4 laboratory found no evidence of aerosol transmission among non-human primates [13].

Marburg haemorrhagic fever (MHF) and Ebola haemorrhagic fever (EHF) should be considered in patients with the appropriate clinical symptoms and exposures, such as recent travel in an endemic area, particularly during seasonal or epidemic disease activity. Clinical suspicion of filovirus infection based on clinical history and presenting symptoms can be supported histopathologically and confirmed by using immunohistochemistry [62].

Gross autopsy findings include petechiae or ecchymoses of the skin or mucous membranes, and haemorrhage of internal organs [63–67]. However, these pathological features are seen in many other diseases and raise a broad differential diagnosis, which includes not only other VHFs but also rickettsial, bacterial, other viral and even non-infectious causes of haemorrhage. More definitive diagnosis can be made by laboratory tests such as PCR, antigen-capture enzyme-linked immunosorbent assay (ELISA), virus isolation, IgM and IgG antibodies, histopathology, IHC and electron microscopy [44,46,47,58].

In this article we review the histopathology, immunohistochemical, *in situ* hybridization and ultrastructural findings of fatal human cases caused by Zaire Ebola virus (EBOV), Sudan virus (SUDV), Bundibugyo virus (BDBV) and Marburg virus (MARV) at the Centers for Disease Control and Prevention during 1995–2007. A total of 89 fatal cases of filoviruses were analysed, from skin biopsies, post-mortem biopsies and autopsies; 74 Ebola virus cases (28 EBOV, 45 SUDV and one BDBV) and 15 Marburg virus cases (14 skin biopsies and one autopsy) were evaluated. A review of the literature identified only 30 fatal cases where an autopsy or post-mortem biopsies were performed and pathological descriptions were reported. This article presents our findings on the 89 cases and compares them with the reports in the literature.

In general, the overall histopathological features are similar in Marburg virus and Ebola virus infections, with

necrosis seen in many organs, including liver, spleen, kidney and gonads, the gastrointestinal tract and endocardium [47,68].

Liver

The liver has the most characteristic histopathological features in filovirus infections. Hepatocyte necrosis ranges from focal to widespread, often with minimal inflammation. Mild to moderate small-droplet steatosis and Kupffer cell hyperplasia are also seen. In EBOV, the portal tracts usually exhibit extensive karyorrhexis and mononuclear inflammatory cell infiltrates (Figure 4A). Hepatocytes may show characteristic intracellular eosinophilic and filamentous or oval virus inclusions, which are predominantly found in periportal zones and surrounding areas of necrosis [44,47,69]. Ultrastructurally, the inclusions are composed of aggregates of viral nucleocapsids (Figure 4B) [44,47,53,58,66,70]. The inclusions are usually more readily identified in EBOV infections than in MARV infections: EBOV infections appeared to contain the most frequent hepatocyte inclusions; SUDV infections demonstrated an intermediate number of inclusions; while no inclusions were identified in a single case of BDBV evaluated (Figures 5–7). Although the literature describes hepatocyte inclusions in Marburg virus infections [58,66,71], we did not observe hepatocyte inclusions in the one case of Marburg virus infection examined (Figure 8A, B).

Lung

Microscopic examination of lung from fatal Ebola cases shows congestion, focal intra-alveolar oedema and haemorrhage with no significant inflammation. Immunohistochemical staining shows viral antigens in alveolar macrophages, endothelial cells, fibroblasts and other interstitial cells [47,62]. By careful review of haematoxylin and eosin (H&E) sections using high-power magnification, viral inclusions can be seen within the intra-alveolar macrophages (Figure 9B) and are more readily identified by using IHC (Figure 9C). Replication of the virus in these macrophages is evidenced by ISH staining (Figure 9D). These inclusions within macrophages can also be seen by EM (Figure 9E). Free viral particles can be seen within the alveolar spaces (Figure 9F). In the few case reports, Marburg viruses do not demonstrate significant lung pathology [65,67]. Only one description was associated with diffuse congestion, haemorrhage and suppurative pneumonia, but the patient had a bacterial co-infection [58]. Experimental studies show variable lesions, which range from oedema, focal haemorrhage and necrosis in non-human primates to pneumonia in pigs [48,72–77].

Spleen and lymph nodes

In EHF, widespread lymphoid depletion associated with necrotic debris and apoptosis have been observed in the spleen and lymph nodes. Viral immunostaining is seen

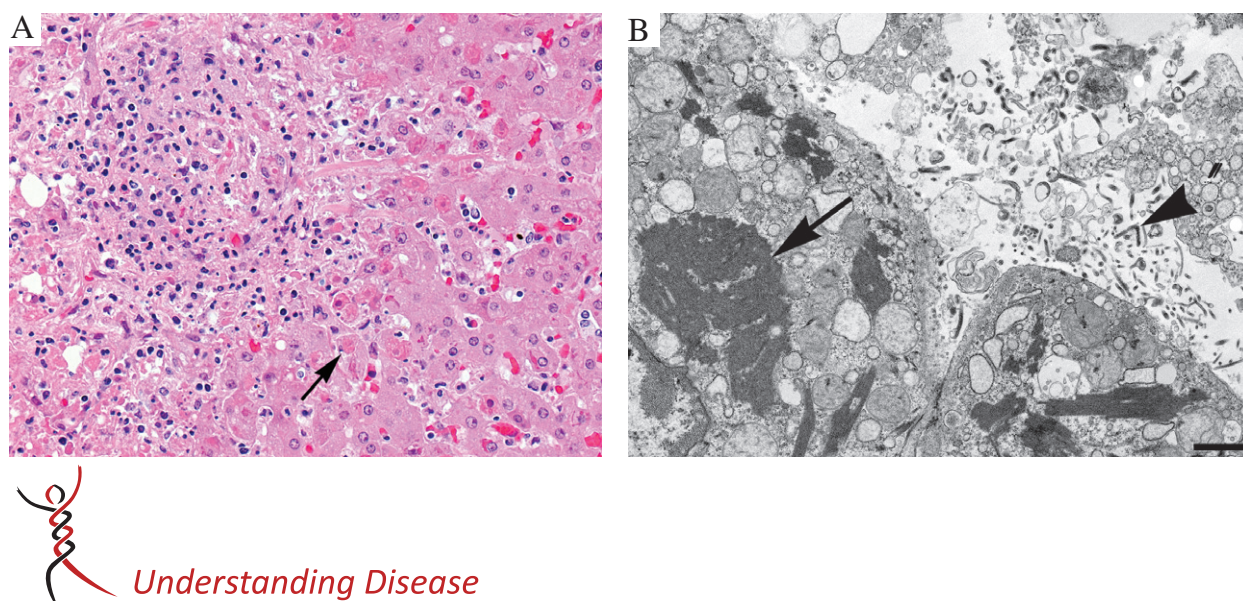


Figure 4. Hepatic histopathological and ultrastructural findings in a fatal case of Ebola virus (EBOV) infection. (A) Low-power photomicrograph showing hepatocellular necrosis, sinusoidal dilatation and congestion; note numerous intracytoplasmic eosinophilic inclusions (arrow). (B) EM photomicrograph shows hepatic viral inclusions (arrow) within infected hepatocytes; note the extracellular sinusoidal virus particles (arrowhead); scale bar = 1 μ m. (A) Haematoxylin and eosin (H&E) stain; (B) thin-sectioned EM, courtesy of Cynthia S Goldsmith, IDPB, CDC

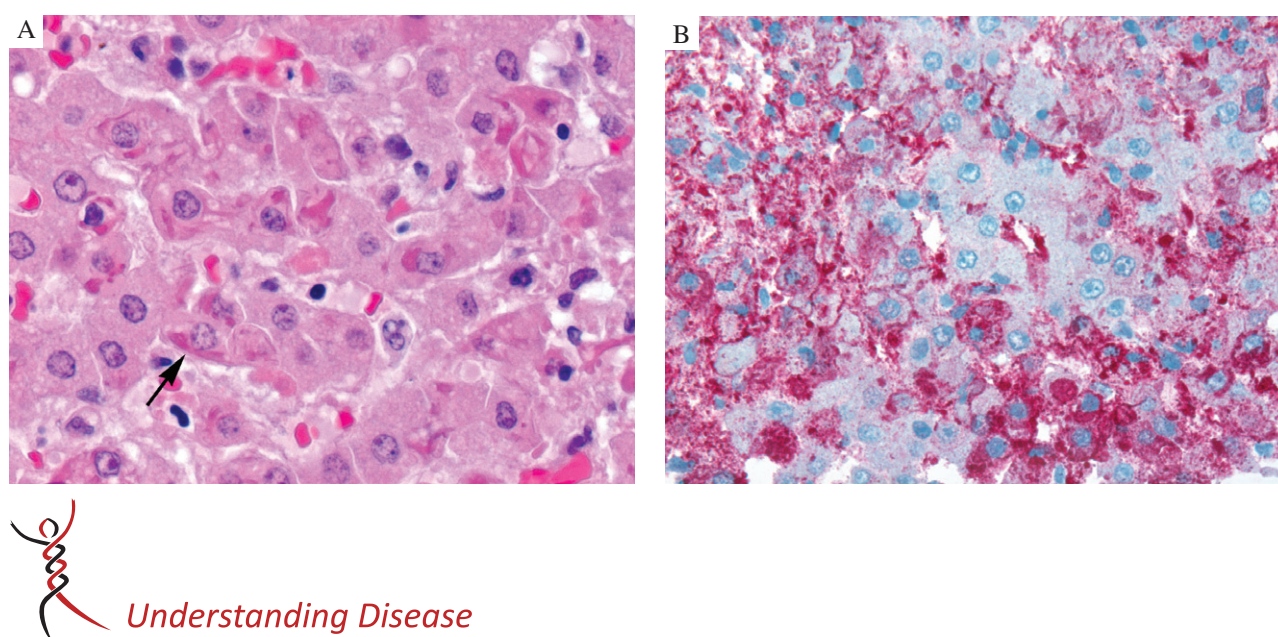


Figure 5. Hepatic histopathological features and viral immunostaining in a fatal case of Ebola virus (EBOV) infection. (A) Hepatocellular necrosis with intracytoplasmic eosinophilic and filamentous inclusions (arrow). (B) Extensive immunostaining of EBOV antigens is seen throughout, in sinusoids, sinusoidal lining cells and hepatocytes. (A) H&E stain; (B) immunoalkaline phosphatase staining, using a rabbit anti-Ebola antibody, naphthol fast red substrate with light haematoxylin counterstain

throughout the spleen, and involves cells of the mononuclear phagocytic system, dendritic cells and fibroblasts (Figure 10A–F) [47,58,65,67]. Additionally, large amounts of antigen are present extracellularly, in association with necrotic cells and debris. Marburg virus infection shows depletion of the white pulp, perifollicular congestion and centrilobular necrosis [66].

Comparisons between the splenic histopathology of different virus strains is extremely difficult, given that

many of the samples are core needle biopsies and limit the ability to assess architectural changes. However, our overall impression is that EBOV infections tend to have greater degrees of lymphoid depletion and necrosis and are found to have comparatively larger amounts of viral antigen by using IHC. BUDV infection appears to show the least amount of lymphoid depletion and antigen, while the changes seen in SUDV cases is somewhat intermediate.

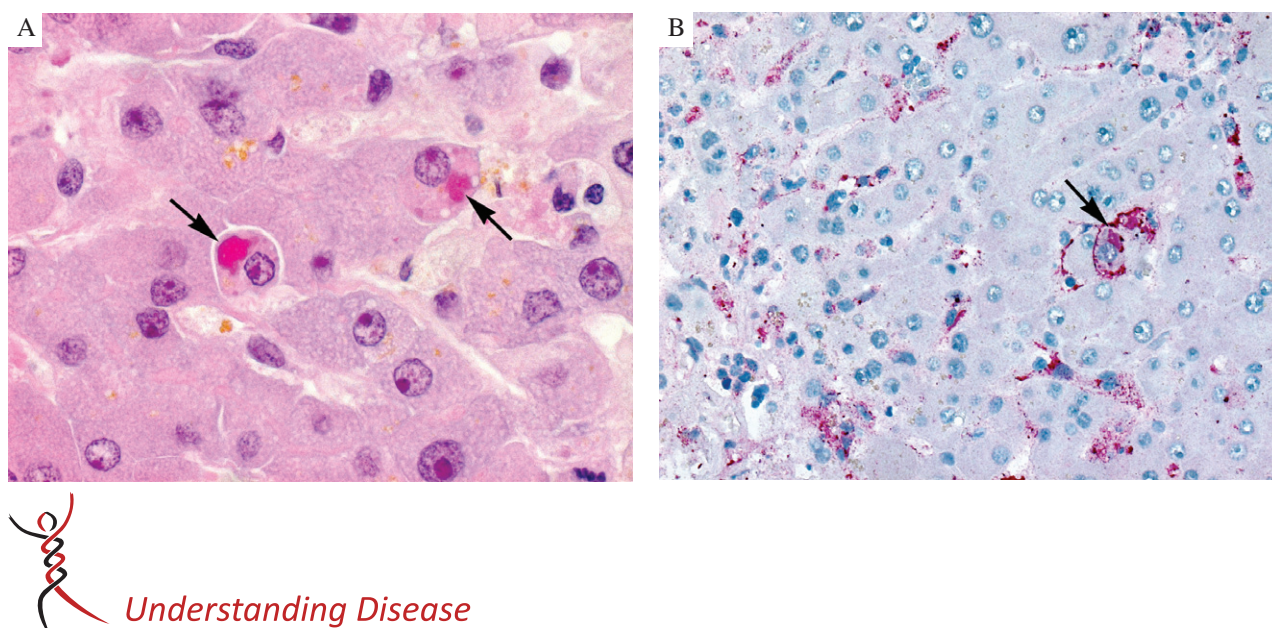


Figure 6. Hepatic histopathological features and viral immunostaining in a fatal case of Sudan virus (SUDV) infection. (A) Rare intracytoplasmic eosinophilic and globular inclusions (arrows). (B) Abundant SUDV antigens are diffusely distributed in Kupffer cells; note the intracytoplasmic immunostaining of an inclusion in a hepatocyte (arrow). (A) H&E stain; (B) immunoalkaline phosphate staining, using a rabbit anti-Ebola antibody, naphthol fast red substrate with light haematoxylin counterstain

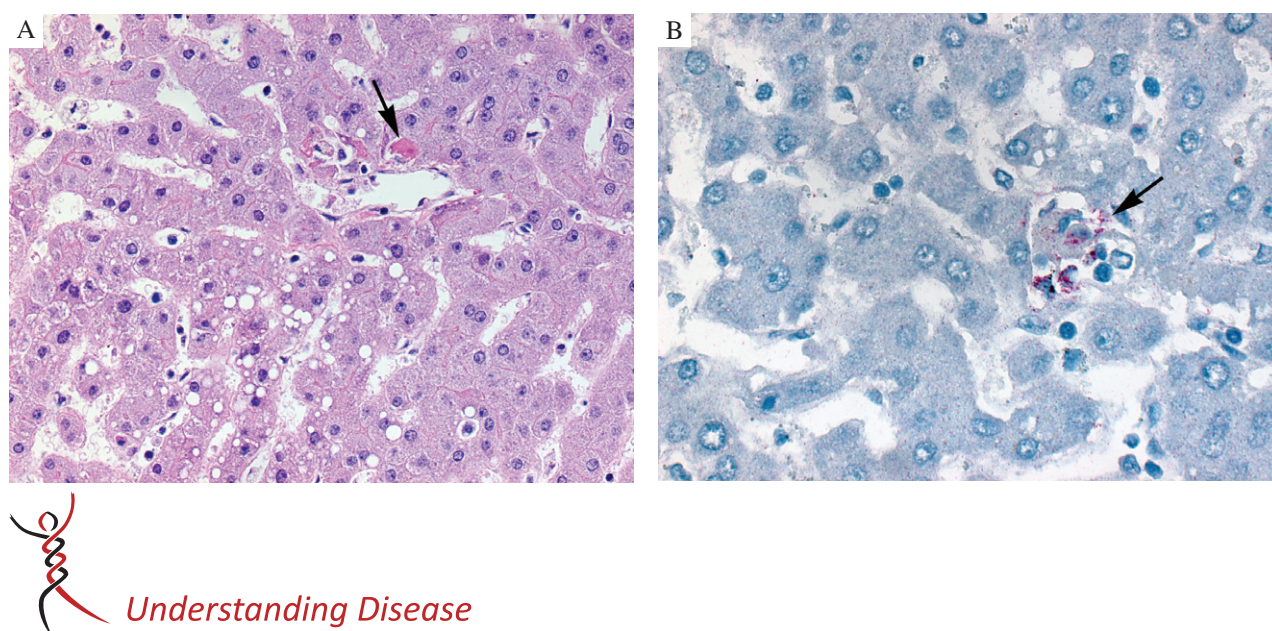


Figure 7. Hepatic histopathological features and viral immunostaining in a fatal case of Bundibugyo virus (BDBV) infection. (A) Small droplet steatosis and focal hepatocyte necrosis (arrow); note the absence of inclusions. (B) BDBV antigens are seen focally (arrow). (A) H&E stain; (B) immunoalkaline phosphate staining, using a rabbit anti-Ebola antibody, naphthol fast red substrate with light haematoxylin counterstain

Skin

In EHF and MHF, signs of haemorrhagic diatheses are commonly seen in the skin and mucous membranes, and are not associated with jaundice. Cutaneous manifestations usually appear 4–5 days after initial symptom onset [78]. The histopathological changes in the skin tissue are minimal and consist mainly of various degrees of dermal oedema, focal haemorrhage and endothelial cell swelling and necrosis. Fibrin

thrombi are rare and vascular necrosis is absent [44,78]. Immunohistochemical stains reveal abundant antigens in epidermal dendritic cells, endothelial cells and connective tissue fibroblasts. Viral antigens can also be detected in the epithelium of sweat and sebaceous glands. Using electron microscopy, viral inclusions and viral particles can be seen within endothelial cells, connective tissue and surface epithelium (Figure 11A–F). Immunohistochemistry on post-mortem skin-punch

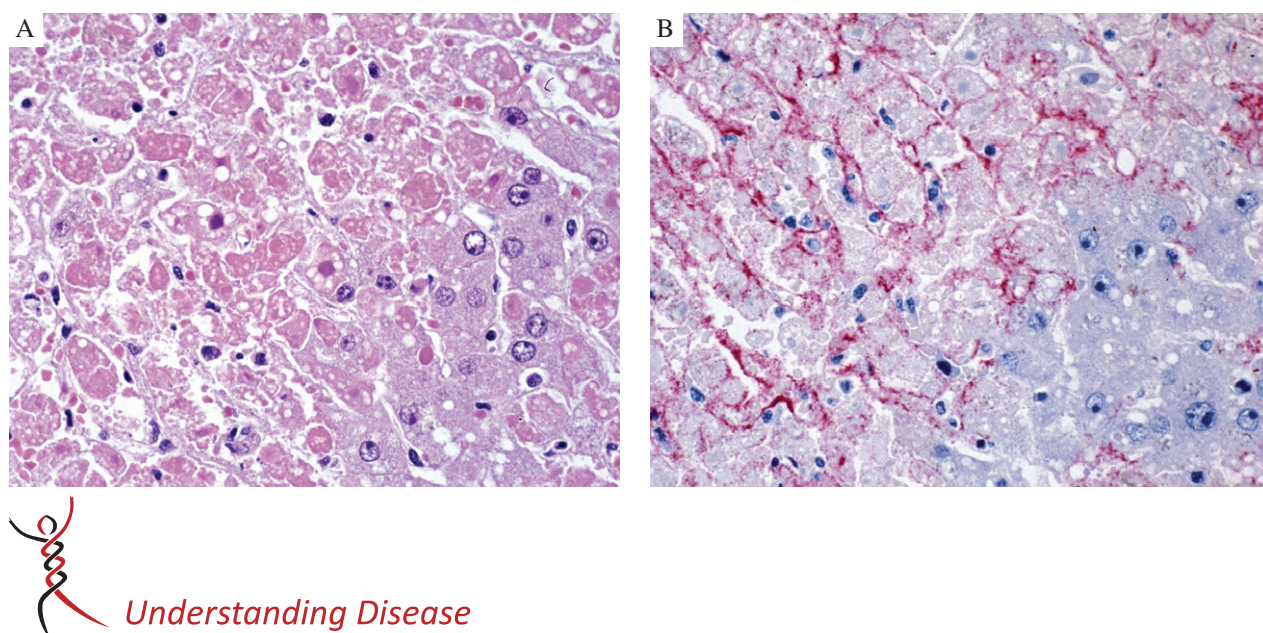


Figure 8. Hepatic histopathological features and viral immunostaining in a fatal case of Marburg virus (MARV) infection. (A) Extensive hepatocellular necrosis adjacent to relatively intact hepatocytes; no distinct viral inclusions are present. (B) Serial sections showing diffuse immunostaining of MARV antigens in sinusoidal lining cells in the necrotic areas observed in (A). (A) H&E stain; (B) immunoalkaline phosphate staining, using a rabbit anti-Ebola antibody, naphthol fast red substrate with light haematoxylin counterstain

biopsies has been used for diagnosis, evaluation of transmission and surveillance [44].

Gastrointestinal tract

In cases of EHF, the gastrointestinal tract shows mild focal mononuclear infiltrates in the lamina propria. Using immunohistochemistry, viral antigens can be seen in mononuclear cells within the lamina propria of gastric, small intestinal and colonic mucosa (Figure 12A, B). Ultrastructurally, Marburg-like inclusions in macrophages and fibroblast-like cells, and virions in necrotic cells and reticular fibres, have been described [58]. These findings correlate with human-to-human transmission resulting from exposure to bloody stools.

Kidney

Kidneys frequently show evidence of acute tubular necrosis with no significant inflammation [47,58]. Immunohistochemistry shows endothelial (including glomerular capillary loops), focal interstitial and tubular staining. Apoptotic debris is frequently seen in the areas of interstitial staining (Figure 13A–D). These findings correlate with studies that found evidence of virus in urine by EM and PCR [79].

Testis

No significant inflammation is observed in the testicular tissue from the one EBOV case examined. Immunohistochemical staining of viral antigens is seen focally within seminiferous tubules, extensively in the endothelium, and in fibrin clots within blood

vessels (Figure 14A–F). Fibrin clots associated with Ebola antigens in the testis and other tissues support the conclusion that thrombocytopenia is likely a result of platelet consumption. Experimental infection in non-human primates shows testicular inflammation and haemorrhage. Electron microscopy shows numerous Ebola virus particles in interstitial cells, endothelium and monocytes [75]. Ebola virus has been detected in semen for up to 82 days in one report, and Marburg virus for up to 13 weeks in another [26,80], after the onset of illness. These findings, along with reported sexual transmission in the latter study, indicate that these viruses can be sexually transmitted [26]. The presence of viral antigens in seminiferous tubules supports the possibility of sexual transmission of filovirus, which has been reported previously.

Bone marrow

The morphological changes in the bone marrow in MHF and EHF are non-specific. The findings include a normocellular bone marrow with focal necrosis. In EBOV cases, abundant inclusions in mononuclear cells, as well as extracellular antigens, can be seen using IHC. Normal megakaryocyte number, and the absence of immunostaining in megakaryocytes, support the conclusion that the thrombocytopenia observed in these infections is not due to reduction in platelet production (Figure 15A, B).

Heart and central nervous system

Morphological myocardial damage in EBOV and MARV autopsy cases is not significant. Immunohistochemistry shows abundant endocardial, endothelial and extracellular antigens (Figure 15C, D). Histological

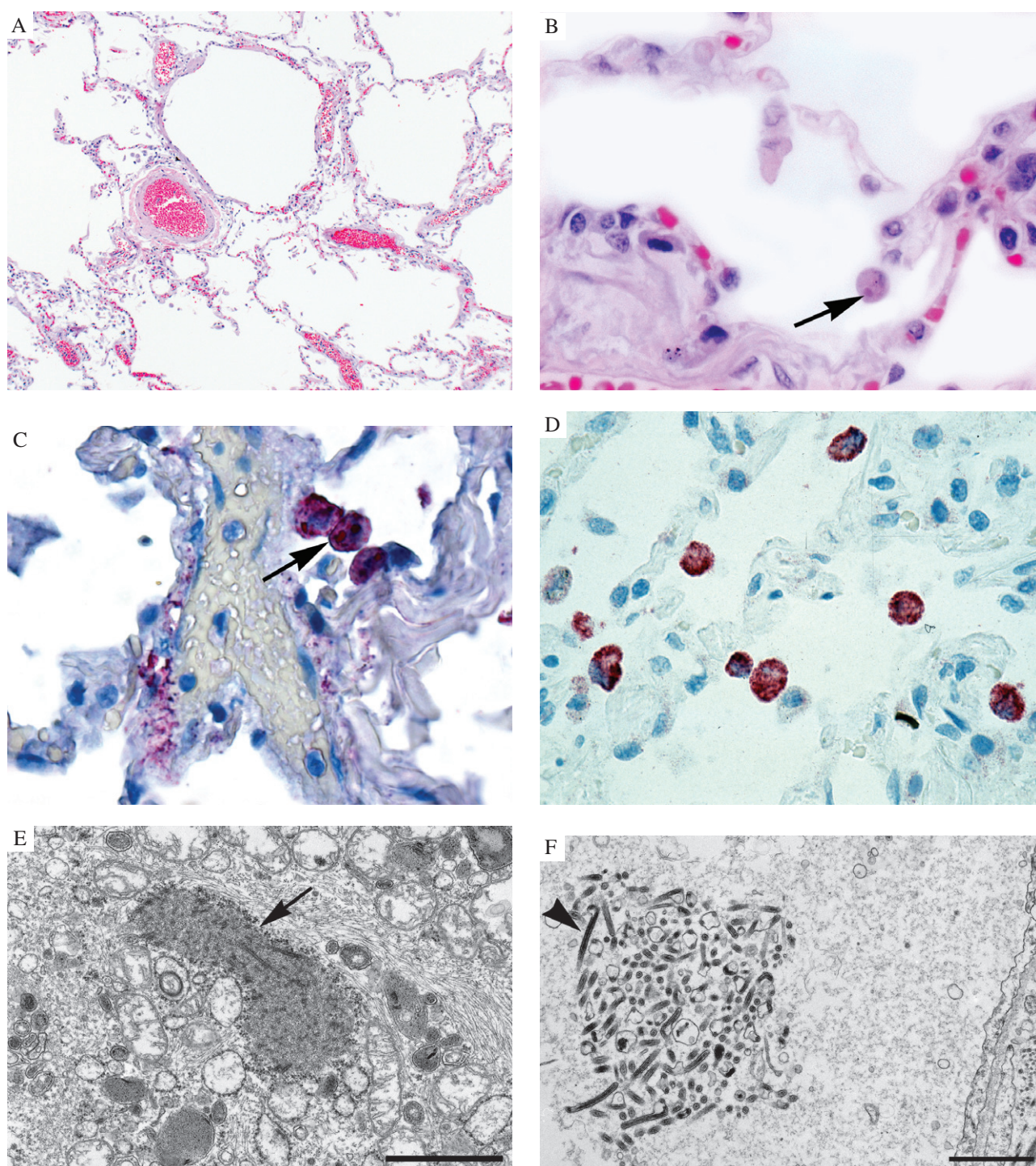


Figure 9. Pulmonary histopathological, immunohistochemical, *in situ* hybridization and ultrastructural characteristics of fatal Ebola haemorrhagic fever. (A) Low-power image of lung shows congestion and absence of significant inflammatory cell infiltrate. (B) High-power image shows an intra-alveolar macrophage with a cytoplasmic eosinophilic inclusion (arrow). (C) Ebola virus antigens in intra-alveolar macrophages and interstitium; note immunostaining of globular inclusions in intra-alveolar mononuclear cells (arrow). (D) *In situ* hybridization, showing evidence of viral replication in intra-alveolar macrophages. (E) EM photomicrograph, showing Ebola virus inclusion (arrow) in an intra-alveolar macrophage, suggesting that viral replication occurs within the macrophage; scale bar = 1 µm. (F) EM photomicrograph, showing extracellular Ebola virus particles (arrowhead) free-floating within the alveolar space; scale bar = 1 µm. (A, B) H&E stain; (C) immunoalkaline phosphate staining, using a rabbit anti-Ebola antibody, naphthol fast red substrate with light haematoxylin counterstain; (D) *in situ* hybridization, using digoxigenin-labelled riboprobe, immunoalkaline phosphate staining and naphthol fast red substrate with light haematoxylin counterstain; (E, F) thin-sectioned EM, courtesy of Cynthia S Goldsmith, IDPB, CDC

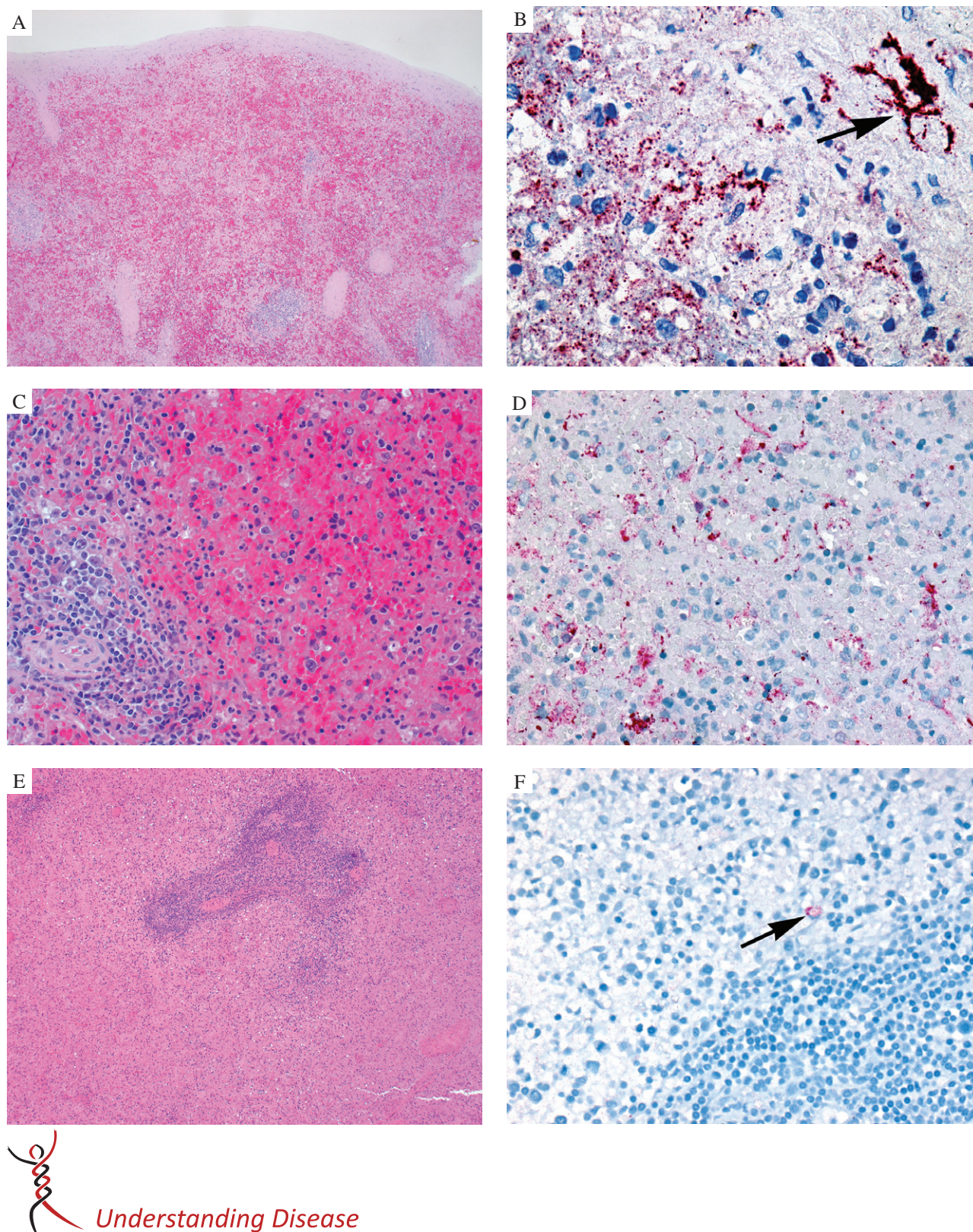


Figure 10. Splenic histopathological features and immunohistochemical characteristics in fatal cases of filovirus infection. Various degrees of lymphoid depletion and congestion are seen in Ebola virus (EBOV), Sudan virus (SUDV) and Bundibugyo virus (BDBV) infection. (A) EBOV case with severe widespread lymphoid depletion and congestion. (B) Immunostaining of Ebola antigens in capsular dendritic cells (arrow) and macrophages in the red pulp. (C) SUDV case, showing a moderate degree of depletion of the germinal centres, haemorrhage and congestion. (D) Immunostaining in macrophages in red pulp associated with karyorrhexis and apoptosis. (E) Low-power photomicrograph of BDBV case, showing a mild degree of lymphoid depletion. (F) Note the rare immunohistochemical staining of a macrophage (arrow) in the red pulp from the same case as in (E). (A, C, E) H&E stain; (B, D, F) immunoalkaline phosphate staining, using a rabbit anti-Ebola antibody, naphthol fast red substrate with light haematoxylin counterstain

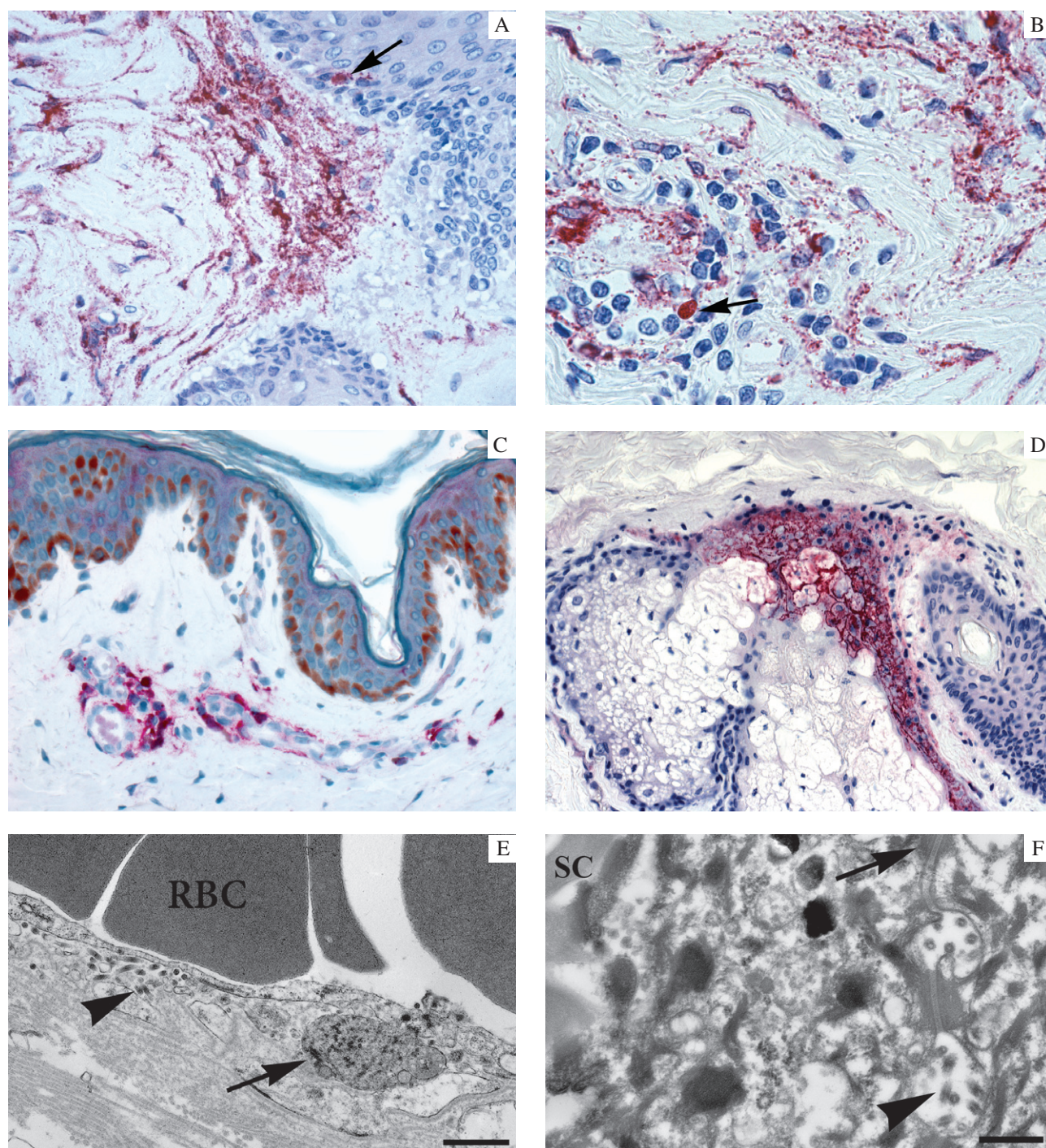


Figure 11. Cutaneous immunohistochemical and ultrastructural features in fatal cases of Ebola virus (EBOV), Sudan virus (SUDV) and Marburg virus (MARV) infection. (A) Extensive amounts of EBOV antigens are seen primarily within fibroblasts in the dermis; note rare immunostaining of a dendritic cell in the surface epithelium (arrow). (B) EBOV immunostaining in connective tissue surrounding sweat glands; note nuclear staining in a sweat gland duct (arrow). (C) SUDV case showing immunostaining of endothelial cells in the superficial dermis. (D) Abundant MARV antigen in sebaceous glands. (E) Ebola virus inclusion (arrow) in an endothelial cell and extracellular virus particles (arrowhead) in dermal collagen; RBC, red blood cell; scale bar = 1 μm . (A–D) Immunoalkaline phosphatase staining, using a rabbit anti-Ebola antibody, naphthol fast red substrate with light haematoxylin counterstain; (E, F) thin-sectioned EM, courtesy of Cynthia S Goldsmith, IDPB. (F) Extracellular Marburg virus particles (arrow head) are seen within spaces between keratinocytes; note desmosomes forming tight junctions (arrow); virus particles are within approximately 3 μm of the stratum corneum (SC)

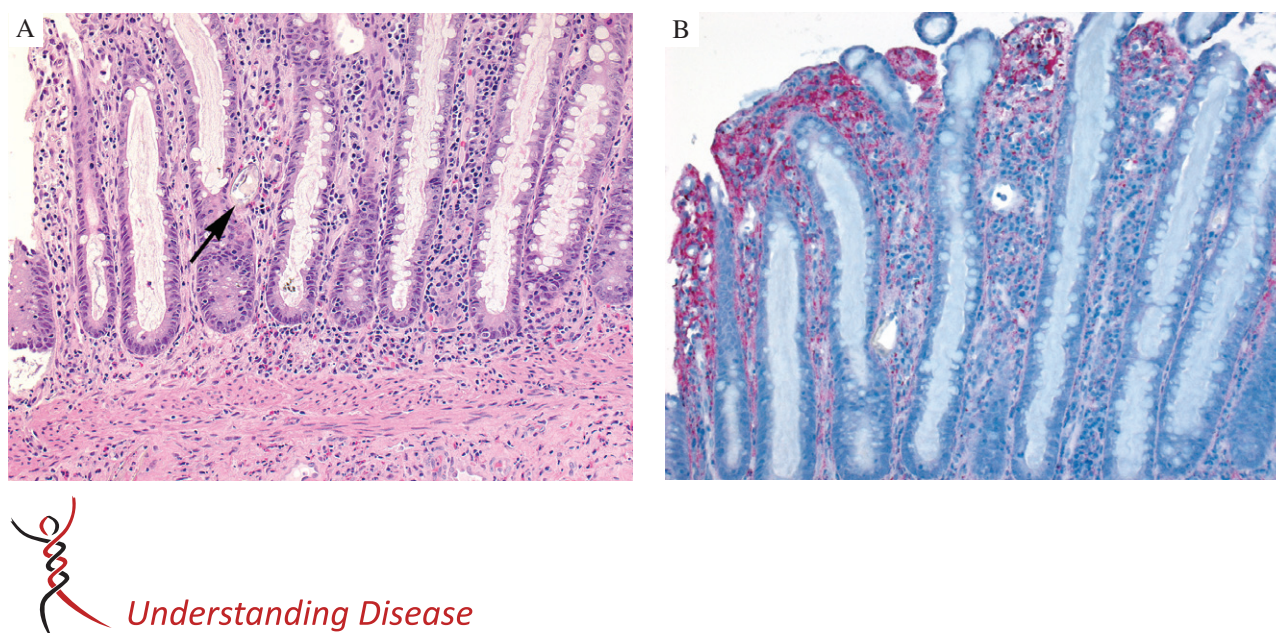


Figure 12. Gastrointestinal tract histopathological features and immunohistochemistry in a fatal case of Ebola haemorrhagic fever. (A) Photomicrograph shows a mild inflammatory infiltrate in the lamina propria of the small intestine; note the incidental schistosoma ovum in the lamina propria (arrow). (B) Immunopositive cells are seen in the lamina propria in the same case as (A). (A) H&E stain; (B) immunoalkaline phosphate staining, using a rabbit anti-Ebola antibody, naphthol fast red substrate with light haematoxylin counterstain

examination of the central nervous system in EHF has never been performed and is limited in MHF, with only two cases reported in the literature. The patients were described to have subacute encephalitis; histologically, the brain showed panencephalitis, with glial nodules and a slight perivascular lymphocytic infiltration [63,67].

Pathogenesis of filoviruses: current hypothesis

In the following sections, the pathogenesis of filovirus infections, including viral entry, early cellular targets, immunopathology, endothelial cell dysfunction and coagulopathy, are discussed (Figure 16).

Filovirus entry

Filoviruses are capable of infecting and replicating in a wide range of tissues and cells. Filoviruses enter the host through mucosal surfaces, breaks and abrasions in the skin or by accidental injection. Filovirus entry involves three distinct phases: cellular attachment, endocytosis and fusion. There are multiple proposed mechanisms of cell entry, which include: clathrin-mediated endocytosis; macropinocytosis [81]; and glycoprotein-facilitated receptor binding [81–85]. Several cellular molecules have been proposed to be receptors or cellular mediators of viral entry, including C-type lectins, tyrosine kinase receptors and, more recently, Niemann Pick C1 [84,86–100]. A wide range of cell types, including hepatocytes and adrenal cortical and medullary cells, have been found to be permissive to filoviruses infection [44,58,101–103]. The interaction of filovirus with many different cellular proteins may explain the broad tropism

seen in filovirus infection. Organ tropism may also be enhanced by direct access of viral particles to the cells of the mononuclear phagocytic system, without the need to penetrate cellular or tissue barriers [104].

Early targets: macrophages and dendritic cells

Macrophages, Kupffer cells and dendritic cells (DCs) have been identified as major early and sustained targets of EBOV and MARV infection [1,47–49,58,105,106]. Moreover, monocytes, macrophages and DCs are thought to be responsible for spreading the virus from the initial site of infection to regional lymph nodes via lymphatics, and to liver and spleen via blood. EBOV subsequently infects tissue macrophages, including Kupffer cells, DCs and fibroblastic reticular cells [105,107].

Immunopathology

The tissue damage seen by histological examination may be interpreted as morphological evidence of the virus's ability to subvert both innate and adaptive immune responses. Despite having high viral load and necrotic lesions in fatal filovirus cases, only minimal inflammation is observed in infected tissues and organs, indicating a dysregulated immune response. Electron microscopic, immunohistochemical and *in situ* hybridization studies have also shown a strong association of parenchymal necrosis in organs and the presence of abundant virus particles, antigens and nucleic acids, with minimal inflammation [44,47,58].

Marburg virus and Ebola virus utilize multiple mechanisms to evade detection and undermine innate immune responses. The structural proteins VP24 and VP35

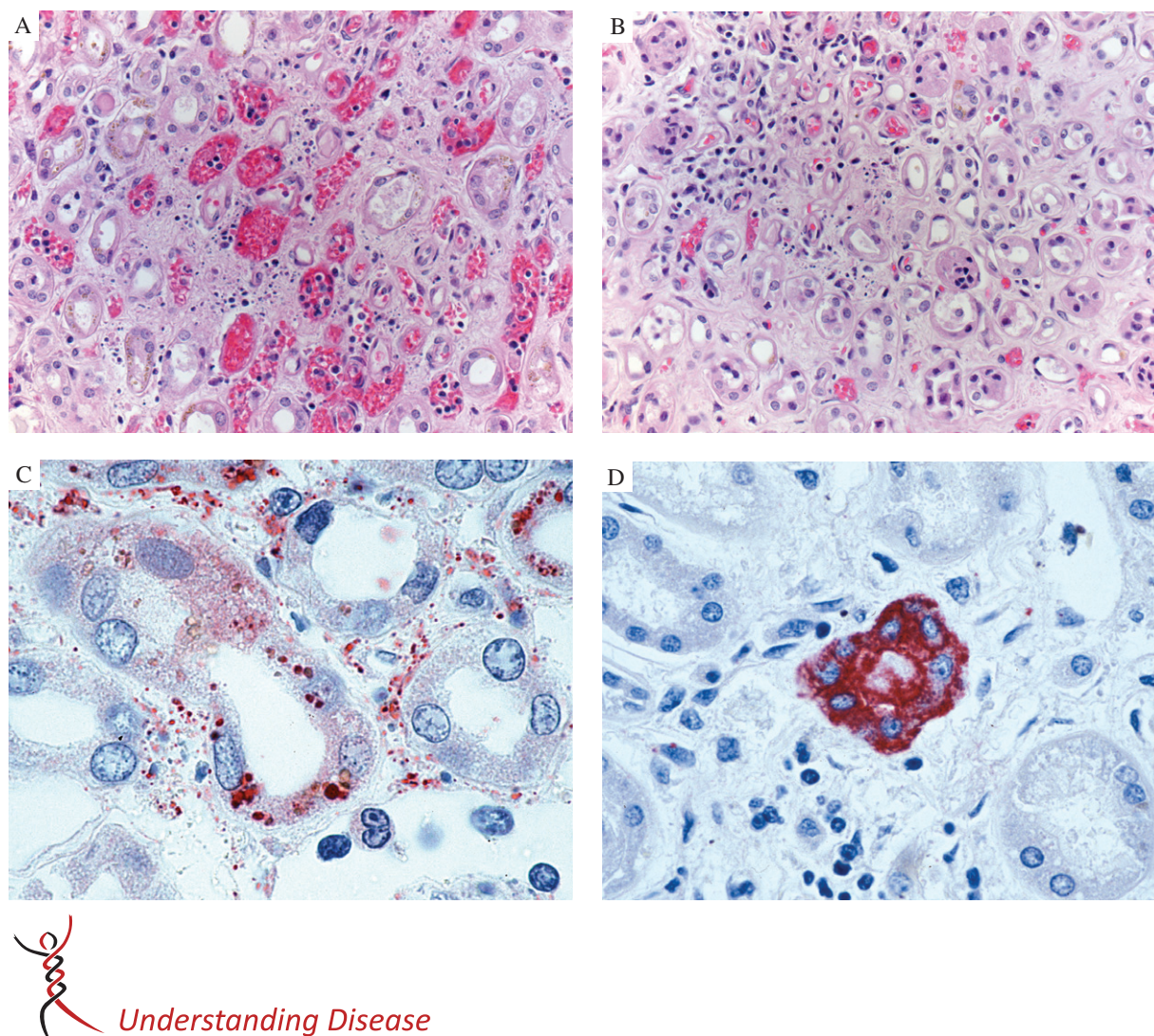


Figure 13. Renal histopathological features and immunohistochemistry in Ebola and Marburg haemorrhagic fevers. (A) Photomicrograph showing vascular congestion and karyorrhectic debris in a fatal case of Ebola virus infection. (B) High-power image showing karyorrhexis and apoptosis. (C) Immunostaining of tubular epithelial cells in association with karyorrhectic debris, in the same case as (A). (D) Focal staining in renal tubular epithelial cells in a fatal case of Marburg virus infection. (A, B) H&E stain; (C, D) immunoalkaline phosphate staining, using a rabbit anti-Ebola antibody, naphthol fast red substrate with light haematoxylin counterstain

are central to EBOV's ability to evade host innate immunity and inhibit type I interferon (IFN) responses [108–117]. The Marburg viruses also diminish host cellular responses to IFNs. The mechanism is thought to be dependent on the structural protein VP40 [118,119].

Several studies have shown that, in cases of severe infection, there is a massive release of pro-inflammatory mediators and vasoactive substances, which promotes inflammation and coagulation but renders the immune system unable to effectively prevent systemic spread of the virus [120–122]. Associations between multiple pro-inflammatory cytokines and increased death rates are seen in human studies [122–125]. Monocytes and macrophages release numerous pro-inflammatory cytokines (IL-1 β , IL-1RA, IL-6, IL-8, IL-15 and IL-16), chemokines and growth factors (MIP-1 α , MIP-1 β , MCP-1, M-CSF, MIF, IP-10, GRO- α and eotaxin). Tumour necrosis factor and reactive oxygen and nitrogen

species are also thought to be involved [119–127]. A recent study noted that the inflammatory response profile in paediatric patients who survived is different from the profile in those who died. Fatal paediatric cases tended to have lower levels of RANTES and higher levels of IL-10 [128]. DCs secrete interleukins and cytokines, which provide a critical link between innate and adaptive immune responses but also act as antigen-presenting cells and initiate adaptive immune responses through T cell activation [106,119,122]. Infected DCs secrete only a limited range of chemokines, fail to express co-stimulatory molecules or up-regulate major histocompatibility complex (MHC) and are unable to induce differentiation of allogenic lymphocytes [84,105,119,128].

Although lymphocytes are not targets of infection, natural killer (NK) cells and T lymphocytes undergo 'bystander' apoptosis relatively early in the course of

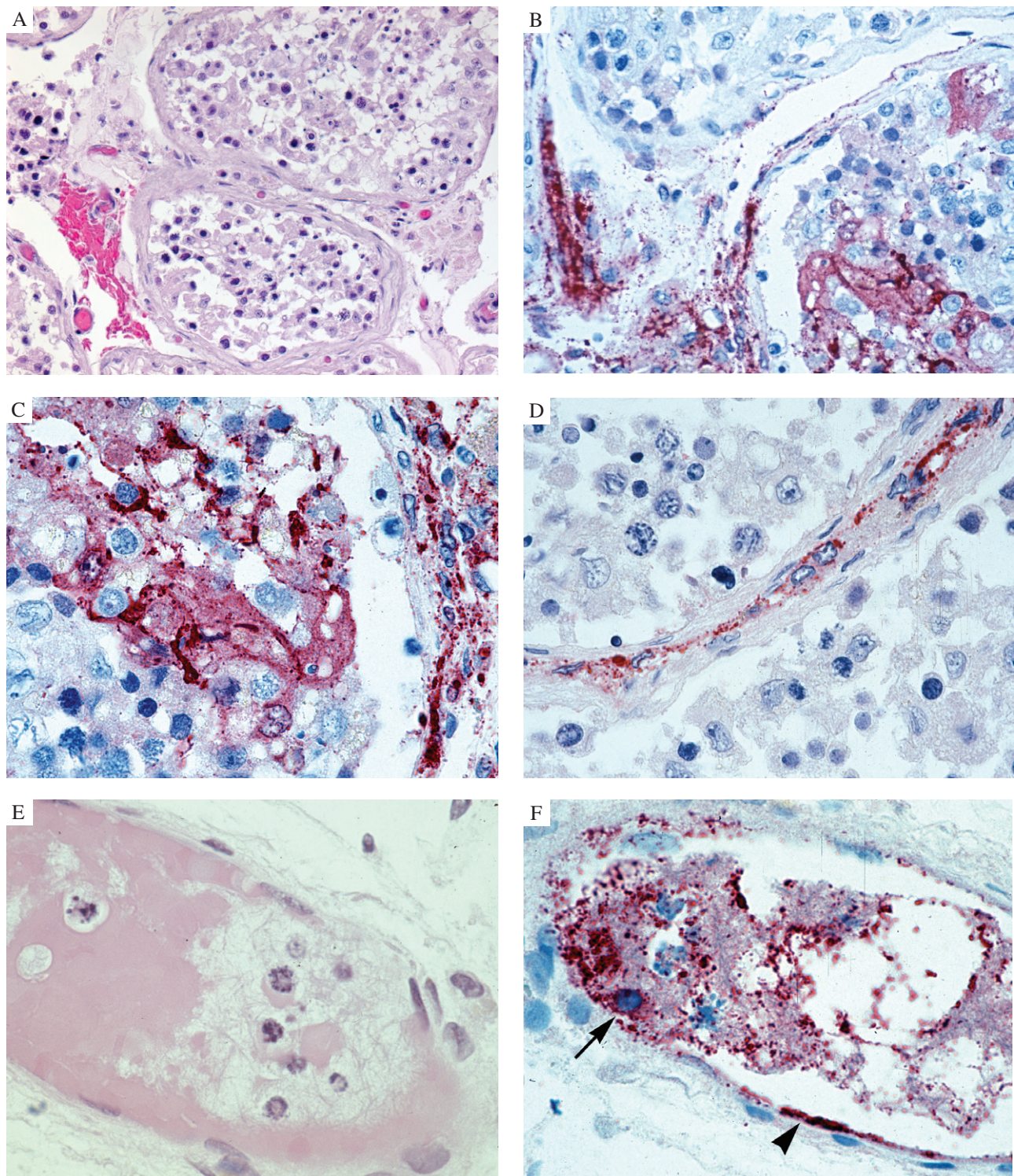


Figure 14. Histopathological and immunohistochemical features in the testis of a fatal case of Ebola virus infection. (A) Photomicrograph showing interstitial congestion and seminiferous tubules with no significant histological changes. (B) Low-power photomicrograph of the same area as (A), showing immunostaining of viral antigens in seminiferous tubules and interstitium, despite the absence of any histological lesions. (C) High-power photomicrograph showing immunostaining of viral antigens in the seminiferous tubules and interstitium. (D) Immunostaining of viral antigens in endothelial cells. (E) Fibrin mesh in a small interstitial blood vessel. (F) Section of the same blood vessel as seen in (E), showing viral immunostaining within fibrin mesh, endothelial cells (arrowhead) and a circulating monocyte (arrow). (A, E) H&E stain; (B, C, D, F) immunoalkaline phosphatase staining, using a rabbit anti-Ebola antibody, naphthol fast red substrate with light haematoxylin counterstain

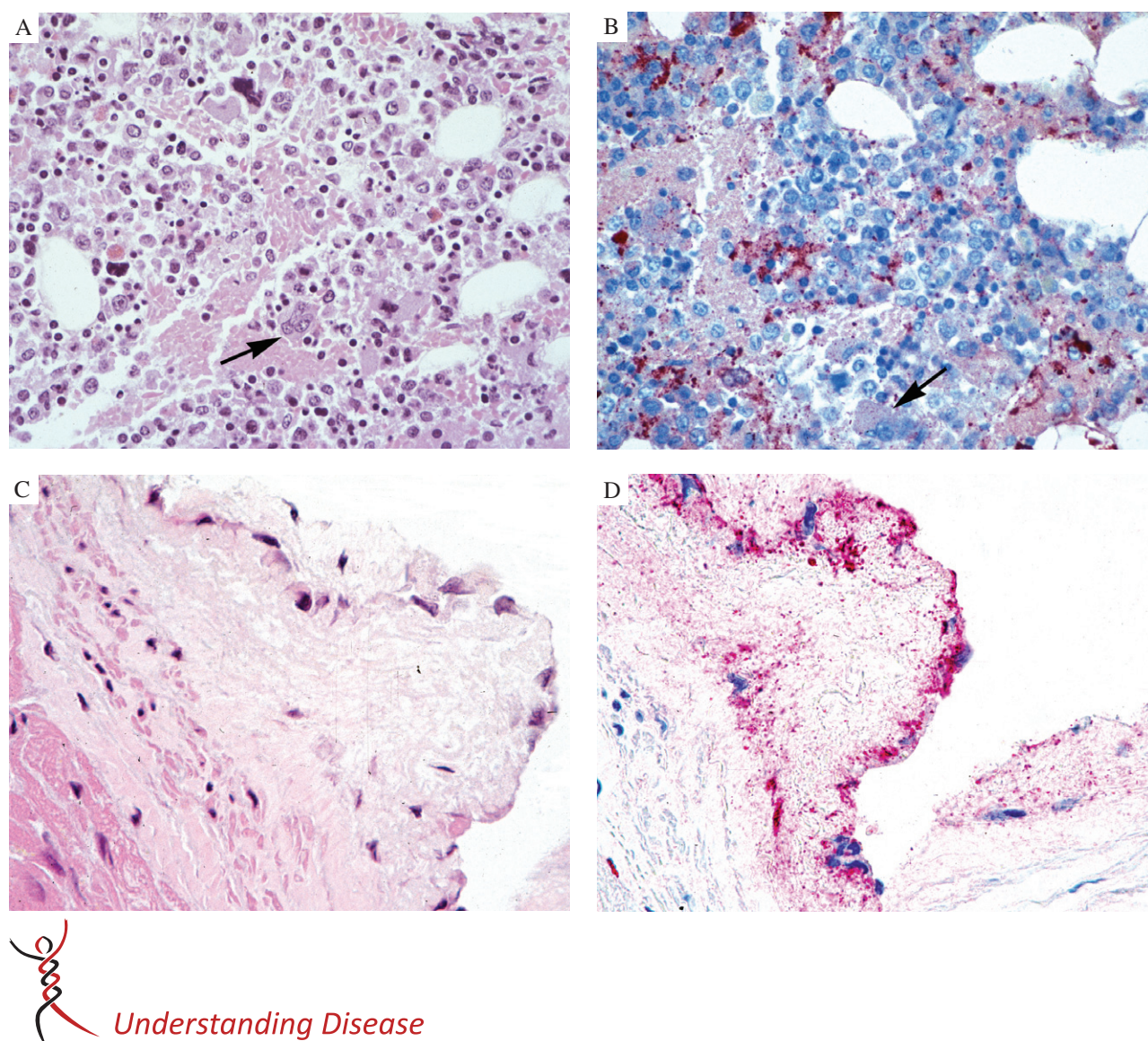


Figure 15. Bone marrow in a fatal case of Ebola virus infection. (A) Normocellular bone marrow with unremarkable megakaryocyte (arrow). (B) Abundant cellular and extracellular staining in the same bone marrow as seen in (A); note the lack of megakaryocyte staining (arrow). (C) Endocardium with no significant histopathology. (D) Abundant immunostaining of Ebola virus antigens in the endocardium and extracellularly in the subendocardium. (A, C) H&E stain; (B, D) immunoalkaline phosphate staining, using a rabbit anti-Ebola antibody, naphthol fast red substrate with light haematoxylin counterstain

the disease [2,122,123,130]. Depletion of NK cells and T lymphocytes may be seen histologically in the spleen and lymph node and further impair the immune response. Infected macrophages and DCs demonstrate increased expression of TNF, Fas and Fas ligand, TNF-related apoptosis-inducing ligand (TRAIL) and nitric oxide (NO), which have all been implicated in inducing apoptosis of host lymphocytes [120,130]. One study observed decreased circulating CD4⁺ and CD8⁺ T lymphocytes, which was associated with increased in Fas in the T lymphocytes, suggesting that the apoptosis is mediated by a Fas–FasL mechanism [123].

Endothelial cell dysfunction

Ultrastructural analyses in fatal cases show numerous inclusions and viral particles within endothelial cells.

Although immunostaining with EBOV antibody is seen in endothelial cells and endocardium, these cells often remain intact and show minimal signs of injury. Cardiac endothelial cell, along with systemic endothelial cell immunostaining and inclusions seen by EM, suggest their involvement in haemorrhage, fluid and electrolyte imbalance and cardiovascular failure. One study showed that endothelial cells in non-human primates are not early targets of EBOV infection [1]. However, abundant EBOV viral particles have been detected in the endothelium of fatal cases [44,47] and may be related to the high viral titres and severity of symptoms seen in these cases. Several studies suggest that endothelial impairment is caused by an indirect immune-mediated mechanism. NO, prostacyclin, IFNs, interleukins and chemokines can modify vascular tone and permeability,

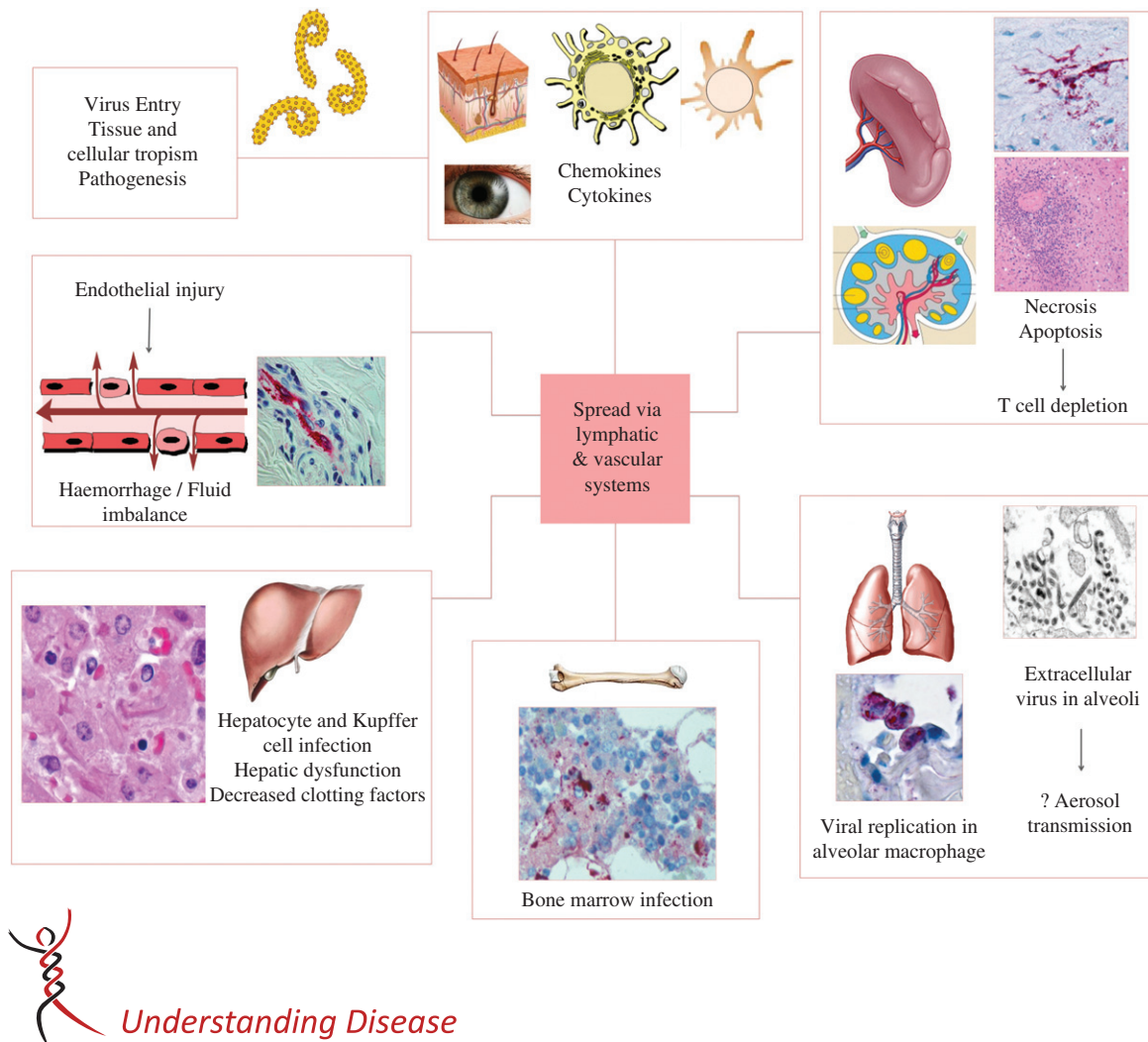


Figure 16. Diagram illustrating filovirus entry, cellular tropism and pathogenesis; M, macrophage; DC, dendritic cell

thrombosis and inflammation, and may contribute to the pathogenesis of hypotensive shock and coagulopathy seen in filovirus infections [131–133].

Coagulopathy

The coagulopathic state seen in EHF is multifactorial and appears to be caused by a combination of activation of the mononuclear phagocytic system, platelet aggregation and consumption, activation of the coagulation cascade, deficiency of coagulation factors due to liver damage, and endothelial damage. Elevated levels of pro-inflammatory cytokines, particularly IL-6, trigger the coagulation cascade [133,134]. Moreover, infected monocytes and macrophages up-regulate the expression of tissue factor (TF) mRNA and the production of cell surface TF protein; one study also detected increased TF-expressing membrane microparticles in macaques infected with EBOV [129,131,133]. Extensive viral replication also causes hepatic necrosis/apoptosis, which further impairs the synthesis of critical coagulation factors. Decreased levels of protein C, protein S and fibrinogen have also been observed [135].

Aberrant activation of both fibrogenic and fibrinolytic pathways leads to disseminated intravascular coagulation (DIC). Markers of fibrinolysis, such as D-dimers, are detectable on day 1 in EBOV-infected non-human primates [129,130].

Treatment and vaccines

The large 2014 outbreak in West Africa highlights that no licensed vaccines or post-exposure treatments against Ebola are available. The first Ebola treatment recently approved by the WHO uses convalescent sera from people who recovered from infection with Ebola virus [136]. Another promising treatment is a combination of monoclonal antibodies (ZMapp), which was tested in rhesus macaques with 100% protection when it was administered as late as 5 days after Ebola exposure [137,138]. These antibodies have already been used on a few patients during this outbreak. The WHO also declared that two candidate vaccines could be introduced as early as this year when safety studies have been concluded [136,139].

Conclusions

Human histopathological studies and ancillary techniques, such as IHC, PCR and EM, are important in the diagnosis of these infections and for enhancing our understanding of their pathogenesis. Additional post-mortem biopsies and autopsy studies will undoubtedly further elucidate the pathology and pathogenesis of filovirus infections in humans.

Haemorrhagic fevers have a complex pathogenesis that involves activation of the mononuclear phagocytic system, with release of pro-inflammatory cytokines, chemokines and growth factors, endothelial dysfunction, alterations of the innate and adaptive immune system, and direct organ and endothelial damage from unrestricted viral replication late in infection and coagulopathy. Viral particles and antigens have been detected in the blood, skin, kidney and seminiferous tubules and explain the role of bodily fluids in human-to-human transmission. Aerosol transmission is extremely controversial, but the finding of free viral particles in alveoli and within intra-alveolar macrophages warrants further study.

Targeted therapies and vaccines are an immediate and top priority. The highly virulent and infectious nature of filoviruses, increasingly porous geographic borders, breakdown of public health measures, poverty and social inequity [140] create significant and daunting challenges ahead. Resolving these issues will require long-term commitment and collaboration from the world community and international partners.

Ethical statement

Tissues used in this study were obtained at autopsy or by post-mortem biopsies and were submitted to the CDC for laboratory evaluation. This activity is not considered research that requires review by an institutional review board or informed consent from patients.

Acknowledgements

The authors acknowledge Dr Wun-Ju Shieh for his assistance in the pathological evaluation of many of the case-patients included in this review. The findings and conclusions are those of the authors and do not necessarily represent the official position of the US Department of Health and Human Services.

Author contributions

All co-authors of this manuscript were responsible for: (a) substantial contributions to conception and design or acquisition of data, or analysis and interpretation of data; (b) drafting the article or revising it critically for important intellectual content; and (c) final approval of the version to be published.

References

- Geisbert TW, Jahrling PB. Exotic emerging viral diseases: progress and challenges. *Nat Med* 2004; **10**: 110–121.
- Paessler S, Walker DH. Pathogenesis of the viral hemorrhagic fevers. *Annu Rev Pathol* 2013; **8**: 411–440.
- Kuhn JH, Becker S, Ebihara H, *et al.* Proposal for a revised taxonomy of the family Filoviridae: classification, names of taxa and viruses, and virus abbreviations. *Arch Virol* 2010; **155**: 2083–2103.
- Hartman AL, Towner JS, Nichol ST. Ebola and Marburg hemorrhagic fever. *Clin Lab Med* 2010; **30**: 161–77.
- Marburg hemorrhagic fever (Marburg HF). Chronology of Marburg hemorrhagic fever outbreaks [Internet]. Centers for Disease Control and Prevention [accessed 13 August 2014]: <http://www.cdc.gov/vhf/marburg/resources/outbreak-table.html>
- Ligon BL. Outbreak of Marburg hemorrhagic fever in Angola: a review of the history of the disease and its biological aspects. *Semin Pediatr Infect Dis* 2005; **16**: 219–224.
- 2014 Ebola outbreak in West Africa (Guinea, Liberia, Sierra Leone and Nigeria) [Internet]. Centers for Disease Control and Prevention [accessed 23 August 2014]: <http://www.cdc.gov/vhf/ebola/outbreaks/guinea/>
- Feldmann H. Ebola – a growing threat? *N Engl J Med* 2014; **371**: 1375–1378.
- Meltzer MI, Atkins CY, Santibanez S, *et al.* Estimating the future number of cases in the Ebola epidemic – Liberia and Sierra Leone, 2014–2015. *Morbidity Mortal Wkly Rep Surveill Summ* 2014; **26**: 1–14.
- Shieh WJ, Hsiao CH, Paddock CD, *et al.* Immunohistochemical, *in situ* hybridization, and ultrastructural localization of SARS-associated coronavirus in lung of a fatal case of severe acute respiratory syndrome in Taiwan. *Hum Pathol* 2005; **36**: 303–309.
- Shieh WJ, Blau DM, Denison AM, *et al.* 2009 pandemic influenza A (H1N1): pathology and pathogenesis of 100 fatal cases in the United States. *Am J Pathol* 2010; **177**: 166–175.
- Zaki S, Blau DM, Hughes JM, *et al.* CDC Grand Rounds: discovering new diseases via enhanced partnership between public health and pathology experts. *Morbidity Mortal Wkly Rep* 2014; **63**: 121–126.
- Alimonti J, Leung A, Jones S, *et al.* Evaluation of transmission risks associated with *in vivo* replication of several high containment pathogens in a biosafety level 4 laboratory. *Sci Rep* 2014; **4**: 5824.
- Siegel JD, Rhinehart E, Jackson M, *et al.* Health Care Infection Control Practices Advisory Committee. 2007 Guideline for isolation precautions: preventing transmission of infectious agents in health care settings. *Am J Infect Control* 2007; **35**: S65–164.
- Interim infection control recommendations for care of patients with suspected or confirmed filovirus (Ebola, Marburg) hemorrhagic fever. BDP/EPR/WHO, Geneva, March 2008 [accessed 6 August 2014]: http://www.who.int/csr/bioriskreduction/interim_recommendations_filovirus.pdf
- Guidance for safe handling of human remains of Ebola patients in US hospitals and mortuaries [Internet]. Centers for Disease Control and Prevention. Ebola hemorrhagic fever. Healthcare workers [accessed 27 August 2014]: <http://www.cdc.gov/vhf/ebola/hcp/guidance-safe-handling-human-remains-ebola-patients-us-hospitals-mortuaries.html>
- Sanchez A, Trappier SG, Mahy BW, *et al.* The virion glycoproteins of Ebola viruses are encoded in two reading frames and are expressed through transcriptional editing. *Proc Natl Acad Sci USA* 1996; **93**: 3602–3607.
- Bukreyev AA, Chandran K, Dolnik O, *et al.* Discussions and decisions of the 2012–2014 International Committee on Taxonomy of Viruses (ICTV) Filoviridae Study Group, January 2012–June 2013. *Arch Virol* 2014; **159**: 821–830.

19. Formenty P, Hatz C, Le Guenno B, *et al.* Human infection due to Ebola virus, subtype Côte d'Ivoire: clinical and biologic presentation. *J Infect Dis* 1999; **179**: S48–53.
20. Pourrut X, Kumulungui B, Wittmann T, *et al.* The natural history of Ebola virus in Africa. *Microbes Infect* 2005; **7**: 1005–1014.
21. Albarino CG, Shoemaker T, Khristova ML, *et al.* Genomic analysis of filoviruses associated with four viral hemorrhagic fever outbreaks in Uganda and the Democratic Republic of the Congo in 2012. *Virology* 2013; **442**: 97–100.
22. Online Marburg virus diseases [Internet]. Global alert and response. World Health Organization [accessed 13 August 2014]: <http://www.who.int/csr/disease/marburg/en/>
23. Nkoghe D, Formenty P, Nnégué S, *et al.* Practical guidelines for the management of Ebola infected patients in the field. *Med Trop* 2004; **64**: 199–204.
24. Smith DH, Johnson BK, Isaacson M, *et al.* Marburg-virus disease in Kenya. *Lancet* 1982; **1**: 816–820.
25. Leroy EM, Kumulungui B, Pourrut X, *et al.* Fruit bats as reservoirs of Ebola virus. *Nature* 2005; **438**: 575–576.
26. Bausch DG, Towner JS, Dowell SF, *et al.* Assessment of the risk of Ebola virus transmission from bodily fluids and fomites. *J Infect Dis* 2007; **196**: 142–147.
27. Piercy TJ, Smither SJ, Steward JA, *et al.* The survival of filoviruses in liquids, on solid substrates and in a dynamic aerosol. *J Appl Microbiol* 2010; **109**: 1531–1539.
28. Nikiforov VV, Turovskii Iu I, Kalinin PP, *et al.* A case of a laboratory infection with Marburg fever. *Zh Mikrobiol Epidemiol Immunobiol* 1994; **3**: 104–106.
29. Brauburger K, Hume AJ, Mühlberger E, *et al.* Forty-five years of Marburg virus research. *Viruses* 2012; **4**: 1878–1927.
30. Biek R, Walsh PD, Leroy EM, Real LA. Recent common ancestry of Ebola Zaire virus found in a bat reservoir. *PLoS Pathog* 2006; **2**: e90.
31. Gatherer D. The 2014 Ebola virus disease outbreak in West Africa. *J Gen Virol* 2014; **95**: 1619–1624.
32. Olival KJ, Islam A, Yu M, *et al.* Ebola virus antibodies in fruit bats, Bangladesh. *Emerg Infect Dis* 2013; **19**: 270–273.
33. Pourrut X, Souris M, Towner JS, *et al.* Large serological survey showing cocirculation of Ebola and Marburg viruses in Gabonese bat populations, and a high seroprevalence of both viruses in *Rousettus aegyptiacus*. *BMC Infect Dis* 2009; **9**: 159.
34. Towner JS, Amman BR, Sealy TK, *et al.* Isolation of genetically diverse Marburg viruses from Egyptian fruit bats. *PLoS Pathog* 2009; **5**: e1000536.
35. Amman BR, Carroll SA, Reed ZD, *et al.* Seasonal pulses of Marburg virus circulation in juvenile *Rousettus aegyptiacus* bats coincide with periods of increased risk of human infection. *PLoS Pathog* 2012; **8**: e1002877.
36. Adjemian J, Farnon EC, Tschioke F, *et al.* Outbreak of Marburg hemorrhagic fever among miners in Kamwenge and Ibanda districts, Uganda, 2007. *J Infect Dis* 2011; **204**: S796–799.
37. Groseth A, Feldmann H, Strong JE. The ecology of Ebola virus. *Trends Microbiol* 2007; **15**: 408–416.
38. Bagcchi S. Ebola haemorrhagic fever in West Africa. *Lancet Infect Dis* 2014; **14**: 375.
39. Barrette RW, Metwally SA, Rowland JM. Discovery of swine as a host for the Reston Ebola virus. *Science* 2009; **10**: 204–206.
40. Dixon MG, Schafer IJ; Centers for Disease Control and Prevention (CDC). Ebola viral disease outbreak – West Africa, 2014. *Morbidity Mortal Wkly Rep* 2014; **63**: 548–551.
41. Kalongi Y, Mwanza K, Tshisuka M, *et al.* Isolated case of Ebola hemorrhagic fever with mucormycosis complications, Kinshasa, Democratic Republic of the Congo. *J Infect Dis* 1999; **179**: S15–17.
42. Bwaka MA, Bonnet MJ, Calain P, *et al.* Ebola hemorrhagic fever in Kikwit, Democratic Republic of the Congo: clinical observations in 103 patients. *J Infect Dis* 1999; **179**: S1–7.
43. Baize S, Leroy EM, Georges-Courbot MC, *et al.* Defective humoral responses and extensive intravascular apoptosis are associated with fatal outcome in Ebola virus-infected patients. *Nat Med* 1999; **5**: 423–426.
44. Zaki SR, Shieh WJ, Greer PW, *et al.* A novel immunohistochemical assay for the detection of Ebola virus in skin: implications for diagnosis, spread, and surveillance of Ebola hemorrhagic fever. Commission de Lutte contre les Epidémies à Kikwit. *J Infect Dis* 1999; **179**: 36–47.
45. Rowe AK, Bertolli J, Khan AS, *et al.* Clinical, virologic, and immunologic follow-up of convalescent Ebola haemorrhagic fever patients and their household contacts, Kikwit, Democratic Republic of the Congo. *J Infect Dis* 1999; **179**: 28–35.
46. Towner JS, Rollin PE, Bausch DG, *et al.* Rapid diagnosis of Ebola hemorrhagic fever by reverse transcription–PCR in an outbreak setting and assessment of patient viral load as a predictor of outcome. *J Virol* 2004; **78**: 4330–4341.
47. Zaki SR, Goldsmith CS. Pathologic features of filovirus infections in humans. *Curr Top Microbiol Immunol* 1999; **235**: 97–116.
48. Connolly BM, Steele KE, Davis KJ, *et al.* Pathogenesis of experimental Ebola virus infection in guinea pigs. *J Infect Dis* 1999; **179**: S203–217.
49. Gibb TR, Bray M, Geisbert TW, *et al.* Pathogenesis of experimental Ebola Zaire virus infection in BALB/c mice. *J Comp Pathol* 2001; **125**: 233–242.
50. Steele K, Crise B, Kuehne A, *et al.* Ebola virus glycoprotein demonstrates differential cellular localization in infected cell types of non-human primates and guinea pigs. *Arch Pathol Lab Med* 2001; **125**: 625–630.
51. Gibb TR, Norwood DA Jr, Woollen N, *et al.* Viral replication and host gene expression in alveolar macrophages infected with Ebola virus (Zaire strain). *Clin Diagn Lab Immunol* 2002; **9**: 19–27.
52. Goldsmith CS, Miller SE. Modern uses of electron microscopy for detection of viruses. *Clin Microbiol Rev* 2009; **22**: 552–563.
53. Bowen ETW, Lloyd G, Harris WJ, *et al.* Viral haemorrhagic fever in southern Sudan and northern Zaire: preliminary studies on the aetiological agent. *Lancet* 1977; **309**: 571–573.
54. Johnson KM, Lange JV, Webb PA, *et al.* Isolation and partial characterisation of a new virus causing acute haemorrhagic fever in Zaire. *Lancet* 1977; **1**: 569–571.
55. Pattyn S, van der Groen G, Courteille G, *et al.* Isolation of Marburg-like virus from a case of haemorrhagic fever in Zaire. *Lancet* 1977; **1**: 573–574.
56. Geisbert TW, Jahrling PB, Hanes MA, *et al.* Association of Ebola-related Reston virus particles and antigen with tissue lesions of monkeys imported to the United States. *J Comp Pathol* 1992; **106**: 137–152.
57. Geisbert TW, Jahrling PB. Differentiation of filoviruses by electron microscopy. *Virus Res* 1995; **39**: 129–150.
58. Geisbert TW, Jaax NK. Marburg hemorrhagic fever: report of a case studied by immunohistochemistry and electron microscopy. *Ultrastruct Pathol* 1998; **22**: 3–17.
59. Le Guenno B, Formenty P, Wyers M, *et al.* Isolation and partial characterization of a new strain of Ebola virus. *Lancet* 1995; **345**: 1271–1274.
60. Nolte KB, Hanzlick RL, Payne DC, *et al.* Medical examiners, coroners, and biologic terrorism: a guidebook for surveillance and case management. *Morbidity Mortal Wkly Rep Recomm Rep* 2004; **53**: 1–27.
61. Nolte KB, Taylor DG, Richmond JY. Biosafety considerations for autopsy. *Am J Forens Med Pathol* 2002; **23**: 107–122.

62. Zaki SR, Paddock CD. Viral diseases. In *Pulmonary Pathology*, 1st edn, Zander DS, Farver CF (eds). Churchill Livingstone Elsevier: Philadelphia, 2008; 245–286.
63. Bechtelsheimer H, Korb G, Gedigk P. The 'Marburg-virus'-hepatitis. Studies in man and guinea pigs. *Virchows Arch A Pathol Pathol Anat* 1970; **351**: 273–290.
64. Ellis DS, Simpson IH, Francis DP, *et al.* Ultrastructure of Ebola virus particles in human liver. *J Clin Pathol* 1978; **31**: 201–208.
65. Gedigk P, Korb G, Bechtelsheimer H. The pathological anatomy of the Marburg virus disease. *Verh Dtsch Ges Pathol* 1968; **52**: 317–320.
66. Rippey JJ, Schepers NJ, Gear JH. The pathology of Marburg virus disease. *S Afr Med J* 1984; **66**: 50–54.
67. Slenczka WG. The Marburg virus outbreak of 1967 and subsequent episodes. *Curr Top Microbiol Immunol* 1999; **235**: 49–75.
68. Zaki SR, Peters CJ. Viral haemorrhagic fevers. In *Pathology of Infectious Diseases*, Connor DH, Chandler FW, Schwartz DA, *et al.* (eds). Appleton and Lange: Stamford, CT, 1997; 347–364.
69. Murphy FA. Pathology of Ebola virus infection. In *Ebola Virus Hemorrhagic Fever*, Pattyn SR (ed.). Elsevier/North-Holland Biomedical: Amsterdam, 1978; 43–59.
70. Lucas SB, Zaki SR, Portmann BC. Other viral and infectious diseases and HIV-related liver disease. In *MacSween's Pathology of the Liver*, 6th edn, Burt AD, Portmann BC, Ferrel LD (eds). Churchill Livingstone Elsevier: London, 2012; 403–466.
71. Bechtelsheimer H, Korb G, Gedigk P. The morphology and pathogenesis of 'Marburg virus' hepatitis. *Human Pathol* 1972; **3**: 255–264.
72. Smither SJ, Nelson M, Eastaugh L, *et al.* Experimental respiratory Marburg virus haemorrhagic fever infection in the common marmoset (*Callithrix jacchus*). *Int J Exp Pathol* 2013; **94**: 156–168.
73. Baskerville A, Bowen ET, Platt GS, *et al.* The pathology of experimental Ebola virus infection in monkeys. *J Pathol* 1978; **125**: 131–138.
74. Ellis DS, Bowen ET, Simpson DI, *et al.* Ebola virus: a comparison, at ultrastructural level, of the behaviour of the Sudan and Zaire strains in monkeys. *Br J Exp Pathol* 1978; **59**: 584–593.
75. Baskerville A, Fisher-Hoch SP, Neild GH, *et al.* Ultrastructural pathology of experimental Ebola haemorrhagic fever virus infection. *J Pathol* 1985; **147**: 199–209.
76. Johnson E, Jaax N, White J, *et al.* Lethal experimental infections of rhesus monkeys by aerosolized Ebola virus. *Int J Exp Pathol* 1995; **76**: 227–236.
77. Nfon CK, Leung A, Smith G, *et al.* Immunopathogenesis of severe acute respiratory disease in Zaire Ebola virus-infected pigs. *PLoS One* 2013; **8**: e61904.
78. Nkoghe D, Leroy EM, Toung-Mve M, *et al.* Cutaneous manifestations of filovirus infections. *Int J Dermatol* 2012; **51**: 1037–1043.
79. Rodriguez LL, De Roo A, Guimard Y. Persistence and genetic stability of Ebola virus during the outbreak in Kikwit, Democratic Republic of the Congo, 1995. *J Infect Dis* 1999; **179**: S170–176.
80. Martini GA, Schmidt HA. Spermatogenic transmission of the 'Marburg virus' (causes of 'Marburg simian disease'). *Klin Wochenschr* 1968; **46**: 398–400.
81. Bhattacharyya S, Mulherkar N, Chandran K. Endocytic pathways involved in filovirus entry: advances, implications and future directions. *Viruses* 2012; **4**: 3647–3664.
82. Takada A, Robison C, Goto H, *et al.* A system for functional analysis of Ebola virus glycoprotein. *Proc Natl Acad Sci USA* 1997; **94**: 14764–14769.
83. Wool-Lewis RJ, Bates P. Characterization of Ebola virus entry by using pseudotyped viruses: identification of receptor-deficient cell lines. *J Virol* 1998; **72**: 3155–3160.
84. Takada A, Kawaoka Y. The pathogenesis of Ebola hemorrhagic fever. *Trends Microbiol* 2001; **9**: 506–511.
85. Hoenen T, Groseth A, Falzarano D, *et al.* Ebola virus: unravelling pathogenesis to combat a deadly disease. *Trends Mol Med* 2006; **12**: 206–215.
86. Sanchez A. Analysis of filovirus entry into vero e6 cells, using inhibitors of endocytosis, endosomal acidification, structural integrity, and cathepsin (B and L) activity. *J Infect Dis* 2007; **196**: S251–258.
87. Takada A. Filovirus tropism: cellular molecules for viral entry. *Front Microbiol* 2012; **3**: 34.
88. Becker S, Spiess M, Klenk HD. The asialoglycoprotein receptor is a potential liver-specific receptor for Marburg virus. *J Gen Virol* 1995; **76**: 393–399.
89. Alvarez CP, Lasala F, Carrillo J, *et al.* C-type lectins DC-SIGN and L-SIGN mediate cellular entry by Ebola virus in *cis* and in *trans*. *J Virol* 2002; **76**: 6841–6844.
90. Lin G, Simmons G, Pöhlmann S, *et al.* Differential N-linked glycosylation of human immunodeficiency virus and Ebola virus envelope glycoproteins modulates interactions with DC-SIGN and DC-SIGNR. *J Virol* 2003; **77**: 1337–1346.
91. Simmons G, Reeves JD, Grogan CC, *et al.* DC-SIGN and DC-SIGNR bind Ebola glycoproteins and enhance infection of macrophages and endothelial cells. *Virology* 2003; **305**: 115–123.
92. Marzi A, Gramberg T, Simmons G, *et al.* DC-SIGN and DC-SIGNR interact with the glycoprotein of Marburg virus and the S protein of severe acute respiratory syndrome coronavirus. *J Virol* 2004; **78**: 12090–12095.
93. Marzi A, Möller P, Hanna SL, *et al.* Analysis of the interaction of Ebola virus glycoprotein with DC-SIGN (dendritic cell-specific intercellular adhesion molecule 3-grabbing nonintegrin) and its homologue DC-SIGNR. *J Infect Dis* 2007; **196**: S237–246.
94. Gramberg T, Soilleux E, Fisch T, *et al.* Interactions of LSECtin and DC-SIGN/DC-SIGNR with viral ligands: differential pH dependence, internalization and virion binding. *Virology* 2008; **373**: 189–201.
95. Takada A, Fujioka K, Tsuiji M, *et al.* Human macrophage C-type lectin specific for galactose and N-acetylgalactosamine promotes filovirus entry. *J Virol* 2004; **78**: 2943–2947.
96. Matsuno K, Kishida N, Usami K, *et al.* Different potential of C-type lectin-mediated entry between Marburg virus strains. *J Virol* 2010; **84**: 5140–5147.
97. Dominguez-Soto A, Aragoneses-Fenoll L, Martin-Gayo E, *et al.* The DC-SIGN-related lectin LSECtin mediates antigen capture and pathogen binding by human myeloid cells. *Blood* 2007; **109**: 5337–5345.
98. Powlesland AS, Fisch T, Taylor ME, *et al.* A novel mechanism for LSECtin binding to Ebola virus surface glycoprotein through truncated glycans. *J Biol Chem* 2008; **283**: 593–602.
99. Carette JE, Raaben M, Wong AC, *et al.* Ebola virus entry requires the cholesterol transporter Niemann–Pick C1. *Nature* 2011; **477**: 340–343.
100. Côté M, Misasi J, Ren T, *et al.* Small molecule inhibitors reveal Niemann–Pick C1 is essential for Ebola virus infection. *Nature* 2011; **477**: 344–348.
101. Hensley LE, Alves DA, Geisbert JB, *et al.* Pathogenesis of Marburg hemorrhagic fever in cynomolgus macaques. *J Infect Dis* 2011; **204**: S1021–1031.
102. Geisbert TW, Hensley LE, Gibb TR, *et al.* Apoptosis induced *in vitro* and *in vivo* during infection by Ebola and Marburg viruses. *Lab Invest* 2000; **80**: 171–186.
103. Geisbert TW, Daddario-DiCaprio KM, Geisbert JB, *et al.* Marburg virus Angola infection of rhesus macaques: pathogenesis and treatment with recombinant nematode anticoagulant protein c2. *J Infect Dis* 2007; **196**: S372–381.

104. Fritz EA, Geisbert JB, Geisbert TW, *et al.* Cellular immune response to Marburg virus infection in cynomolgus macaques. *Viral Immunol* 2008; **21**: 355–363.
105. Schnittler HJ, Feldmann H. Marburg and Ebola hemorrhagic fevers: does the primary course of infection depend on the accessibility of organ-specific macrophages? *Clin Infect Dis* 1998; **27**: 404–406.
106. Geisbert TW, Hensley LE, Larsen T, *et al.* Pathogenesis of Ebola hemorrhagic fever in cynomolgus macaques: evidence that dendritic cells are early and sustained targets of infection. *Am J Pathol* 2003; **163**: 2347–2370.
107. Twenhafel NA, Mattix ME, Johnson JC, *et al.* Pathology of experimental aerosol Zaire Ebola virus infection in rhesus macaques. *Vet Pathol* 2013; **50**: 514–529.
108. Leung DW, Prins KC, Basler CF, *et al.* Ebola virus VP35 is a multifunctional virulence factor. *Virulence* 2010; **1**: 526–531.
109. Zhang AP, Abelson DM, Bornholdt ZA, *et al.* The Ebola virus VP24 interferon antagonist: know your enemy. *Virulence* 2012; **3**: 440–445.
110. Lubaki NM, Ilinykh P, Pietzsch C, *et al.* The lack of maturation of Ebola virus-infected dendritic cells results from the cooperative effect of at least two viral domains. *J Virol* 2013; **87**: 7471–7485.
111. Xu W, Edwards MR, Borek DM, *et al.* Ebola virus VP24 targets a unique NLS binding site on karyopherin- $\alpha 5$ to selectively compete with nuclear import of phosphorylated STAT1. *Cell Host Microbe* 2014; **16**: 187–200.
112. Reid SP, Leung LW, Hartman AL, *et al.* Ebola virus VP24 binds karyopherin $\alpha 1$ and blocks STAT1 nuclear accumulation. *J Virol* 2006; **80**: 5156–5167.
113. Halfmann P, Neumann G, Kawaoka Y. The Ebola virus VP24 protein blocks phosphorylation of p38 mitogen-activated protein kinase. *J Infect Dis* 2011; **204**: S953–956.
114. Zhang AP, Bornholdt ZA, Liu T, *et al.* The Ebola virus interferon antagonist VP24 directly binds STAT1 and has a novel, pyramidal fold. *PLoS Pathog* 2012; **8**: e1002550.
115. Prins KC, Cárdenas WB, Basler CF. Ebola virus protein VP35 impairs the function of interferon regulatory factor-activating kinases IKK ϵ and TBK-1. *J Virol* 2009; **83**: 3069–3077.
116. Ramanan P, Shabman RS, Brown CS, *et al.* Filoviral immune evasion mechanisms. *Viruses* 2011; **3**: 1634–1649.
117. Chang TH, Kubota T, Matsuoka M, *et al.* Ebola Zaire virus blocks type I interferon production by exploiting the host SUMO modification machinery. *PLoS Pathog* 2009; **5**: e1000493.
118. Basler CF, Amarasinghe GK. Evasion of interferon responses by Ebola and Marburg viruses. *J Interferon Cytokine Res* 2009; **29**: 511–520.
119. Valmas C, Grosch MN, Schumann M, *et al.* Marburg virus evades interferon responses by a mechanism distinct from Ebola virus. *PLoS Pathog* 2010; **6**: e1000721.
120. Bray M, Geisbert TW. Ebola virus: the role of macrophages and dendritic cells in the pathogenesis of Ebola hemorrhagic fever. *Int J Biochem Cell Biol* 2005; **37**: 1560–1566.
121. Feldmann H, Bugany H, Mahner F, *et al.* Filovirus-induced endothelial leakage triggered by infected monocytes/macrophages. *J Virol* 1996; **70**: 2208–2214.
122. Zampieri CA, Sullivan NJ, Nabel GJ. Immunopathology of highly virulent pathogens: insights from Ebola virus. *Nat Immunol* 2007; **8**: 1159–1164.
123. Wauquier N, Becquart P, Padilla C, *et al.* Human fatal Zaire Ebola virus infection is associated with an aberrant innate immunity and with massive lymphocyte apoptosis. *PLoS Negl Trop Dis* 2010; **4**: e837.
124. Feldmann H, Geisbert TW. Ebola haemorrhagic fever. *Lancet* 2011; **377**: 849–862.
125. Villinger F, Rollin PE, Brar SS, *et al.* Markedly elevated levels of interferon (IFN)- γ , IFN- α , interleukin (IL)-2, IL-10, and tumor necrosis factor- α associated with fatal Ebola virus infection. *J Infect Dis* 1999; **179**: S188–191.
126. Baize S, Leroy EM, Georges AJ, *et al.* Inflammatory responses in Ebola virus-infected patients. *Clin Exp Immunol* 2002; **128**: 163–168.
127. Hutchinson KL, Rollin PE. Cytokine and chemokine expression in humans infected with Sudan Ebola virus. *J Infect Dis* 2007; **196**: S357–363.
128. McElroy AK, Erickson BR, Flietstra TD, *et al.* Ebola hemorrhagic fever: novel biomarker correlates of clinical outcome. *J Infect Dis* 2014; **210**: 558–566.
129. Mahanty S, Hutchinson K, Agarwal S, *et al.* Cutting edge: impairment of dendritic cells and adaptive immunity by Ebola and Lassa viruses. *J Immunol* 2003; **170**: 2797–2801.
130. Gupta M, Spiropoulou C, Rollin PE. Ebola virus infection of human PBMCs causes massive death of macrophages, CD4 and CD8 T cell sub-populations *in vitro*. *Virology* 2007; **364**: 45–54.
131. Geisbert TW, Young HA, Jahrling PB, *et al.* Pathogenesis of Ebola hemorrhagic fever in primate models: evidence that hemorrhage is not a direct effect of virus-induced cytolysis of endothelial cells. *Am J Pathol* 2003; **163**: 2371–2382.
132. Hensley LE, Young HA, Jahrling PB, *et al.* Proinflammatory response during Ebola virus infection of primate models: possible involvement of the tumor necrosis factor receptor superfamily. *Immunol Lett* 2002; **80**: 169–179.
133. Hensley LE, Geisbert TW. The contribution of the endothelium to the development of coagulation disorders that characterize Ebola hemorrhagic fever in primates. *Thromb Haemost* 2005; **94**: 254–261.
134. Leroy EM, Baize S, Volchkov VE, *et al.* Human asymptomatic Ebola infection and strong inflammatory response. *Lancet* 2000; **355**: 2210–2215.
135. Ebihara H, Rockx B, Marzi A, *et al.* Host response dynamics following lethal infection of rhesus macaques with Zaire Ebola virus. *J Infect Dis* 2011; **204**: S991–999.
136. Gulland A. First Ebola treatment is approved by WHO. *Br Med J* 2014; **349**: g 5539; DOI: 10.1136/bmj.g5539.
137. Qiu X, Wong G, Audet J, Bello A, *et al.* Reversion of advanced Ebola virus disease in nonhuman primates with ZMapp. *Nature* 2014; **514**: 12703–12714.
138. Geisbert TW. Medical research: Ebola therapy protects severely ill monkeys. *Nature* 2014; **514**: 41–43.
139. Cohen J. Infectious Disease. Ebola vaccines racing forward at record pace. *Science* 2014; **345**: 1228–1229.
140. Online microbial threats to health: emergence, detection, and response. Institute of Medicine, 2003: <http://iom.edu/~media/Files/Report%20Files/2003/Microbial-Threats-to-Health-Emergence-Detection-and-Response/MicrobialThreat8pgFINAL.pdf>